The Political Economy of Conservation

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Abstract

Conservation is political. Forest protections safeguard our natural forest resources, but they also provoke resistance from agricultural producers who profit from clearing land. We study this tension in the Brazilian Amazon. Using detailed data on campaign donations and candidates in state and federal elections from 2002 to 2018, we show that environmental reforms led agricultural producers to sharply increase political support for pro-agricultural politicians. Donations from agricultural firm owners roughly doubled, and agricultural candidates gained 70 to 90 percent more votes. Motivated by this evidence, we develop a model of optimal regulation under the threat of political repeal. Regulation that imposes large losses on producers provokes resistance, which leads to repeal that can undo the welfare gains from conservation. We estimate the model with spatial data on land use and agricultural returns across the Brazilian Amazon from 2001 to 2019. Pigouvian taxation maximizes welfare in isolation, but it triggers certain repeal and thus achieves little in expectation. Optimal policy softens taxation and targets sites where conservation gains are large relative to producer losses. This targeted approach achieves substantial feasible welfare gains where Pigouvian taxation cannot.

1 Introduction

Agricultural production drives deforestation that accounts for one fifth of global carbon emissions. This negative externality motivates efforts to strengthen forest regulation, but such regulation faces political headwinds. Large agricultural producers derive sizeable profits from deforestation and thus face strong incentives to lobby against regulation. How can we design conservation policy that accounts for this political reality?

We tackle this question in the Brazilian Amazon. In 2004, the Brazilian government launched the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm), which strengthened enforcement of existing forest protections through satellite monitoring, expanded protected areas, and tougher penalties for illegal clearing. Deforestation fell sharply. We compile detailed data on individual campaign donors and candidates in state and federal elections from 2002 to 2018, linking them to firm-level records from the Brazilian tax authority to identify agricultural sector participants.

We use a difference-in-differences strategy that compares agricultural donors and candidates to their non-agricultural counterparts before and after the implementation of PPCDAm. The identifying assumption is that, absent the reform, political contributions by agricultural firm owners would have evolved in parallel with those of other firm owners. We support this assumption with pre-treatment event studies and present robustness checks using matched samples, as well as triple-difference specifications that exploit variation in enforcement intensity across municipalities and that compare the Legal Amazon to the rest of Brazil.

We find that PPCDAm led to large increases in donations from agricultural donors, particularly to candidates with ties to the agricultural sector. Donations from agricultural firm owners roughly doubled relative to pre-reform levels. Triple-difference estimates confirm that the increase was concentrated where enforcement was strongest. Switching to the candidate side, we document that agricultural candidates received 70 to 90 percent more votes in both state and federal elections, with meaningful positive effects on the probability of being elected to office. These results suggest deliberate efforts to back politicians who oppose environmental regulation. The agricultural sector met regulation with resistance.

Such resistance creates a trade-off for the regulator. We characterize this trade-off

with a simple theory framework that emphasizes the incidence of regulation and the threat of repeal. The typical theoretical approach is instead to maximize aggregate welfare, while addressing distributional impacts with lump-sum transfers. We document political resistance to regulation that is in conflict with the idea that transfers can navigate politics. We take seriously that regulation imposes large losses on agricultural producers, provoking resistance that risks undoing the welfare gains from conservation.

The regulator must therefore balance the environmental benefits of strong regulation against the risk of repeal. We show that optimal regulation falls short of the Pigouvian benchmark, and that precise targeting can ease the trade-off. Regulation that targets sites with high emissions and low agricultural values achieves large welfare gains with limited producer losses, reducing the risk of repeal. By contrast, blunt regulation that burdens all producers, including those on efficient land, invites resistance without meaningful environmental benefit. At the same time, such targeting requires the regulator to observe private values that are typically known only to producers. The gains from targeting depend on the quality of this information.

We study these forces quantitatively with a simple empirical model of agricultural land use in the Brazilian Amazon. We compile spatial data on agricultural production and suitability across the Amazon for the period from 2001 to 2019. The data reveal substantial heterogeneity in both agricultural returns and carbon stocks. We estimate a logit model of land allocation that recovers the responsiveness of the agricultural frontier to changes in profitability. Agricultural returns combine spatial variation in agricultural suitability with time variation in global commodity prices. Estimation reduces to a Poisson regression that we estimate in pooled long differences over 10-year horizons, as well as in annual panel regressions. Our baseline analysis relies on the long-difference specification, which isolates longer-term, decadal responses. By contrast, the panel regression isolates short-run responses to year-on-year fluctuations, and it delivers estimates that are substantially smaller. We use the estimated model to simulate counterfactual regulation, computing welfare, producer surplus, emissions, and the risk of political repeal under alternative policy instruments.

Our counterfactual analysis yields three main findings. First, Pigouvian taxation maximizes welfare in the absence of political resistance, but it also imposes large burdens on producers. This heavy burden prompts political resistance that can trigger repeal. In expectation, Pigouvian taxation achieves little when strong political

resistance guarantees repeal. Second, optimal policy softens taxation and targets regulation toward sites where emissions reductions are large relative to producer losses. This two-dimensional targeting – on both emissions and profits – achieves substantial feasible welfare gains where Pigouvian taxation cannot. Third, policies constrained to target emissions alone, including regional taxes and protected areas, can still effectively navigate the trade-off between welfare and repeal. These constrained policies dominate Pigouvian taxation, particularly when the value of abatement is high, even though they fall short of optimal policy. The gap between optimal and constrained policy reflects the informational challenge of targeting on profits, which are typically the private information of the firm.

This paper contributes to work at the intersection of political economy and the environment (Oates and Portney 2003, Besley and Persson 2023). A growing literature studies how political forces shape climate and environmental outcomes, including climate attitudes and public opinion (Dechezleprêtre et al. 2025, Hsiao and Kuipers 2025), environmental issues in politics (Bombardini et al. 2025, Longuet-Marx 2025), climate adaptation (Hsiao 2025), green investment (Kashner 2026), policy lobbying (Kang 2016), regulatory incidence (Sallee 2022), and regulation of the commons (Ryan and Sudarshan 2022, Berman and Hickok 2026). We bring this perspective to tropical deforestation, where the tension between economic development and environmental protection is acute and the political constraints on regulation are binding.

A large empirical literature studies tropical deforestation in Brazil and Indonesia, which house two of the world's most important tropical ecosystems.[1] In Brazil, recent work has used a range of empirical methods to study the role of policy in shaping land use (Souza-Rodrigues 2019, Assunção et al. 2024, Leite-Mariante and Restrepo 2024, Araujo et al. 2026, Domínguez-Iino 2026). Several papers evaluate the PPCDAm reform, as well as related enforcement and financial instruments (Nepstad et al. 2009, Hargrave and Kis-Katos 2013, Assunção et al. 2015, 2019, Assunção et al. 2022). In Indonesia, the expansion of palm oil cultivation has sparked forest fires and destroyed peatland (Busch et al. 2015, Balboni et al. 2025, Hsiao 2026). More broadly, agricultural expansion drives extensive deforestation at global scale (Hsiao et al. 2026). We document that regulation induces political resistance, which can potentially reverse the initial gains from forest protection.

A smaller set of papers focuses specifically on the politics of deforestation. Glob-

[1] Recent surveys include Balboni et al. (2023) and Costa et al. (2025a).

ally, electoral cycles shape deforestation through illegal logging, forest fires, and weakened enforcement (Burgess et al. 2012, Balboni et al. 2021, Sanford 2023, Harding et al. 2024). In Brazil, agricultural interests and political interference have eroded conservation gains (Burgess et al. 2025, Costa et al. 2025b, Dahis and Katovich 2025). In this crucial setting, we quantify how politics shape the effects and optimal design of environmental regulation. We emphasize the need to account for political sustainability in evaluating policy impacts. The future of the Amazon depends on durable conservation in the face of opposition.

2 The Brazilian Amazon

We describe our institutional setting and data. Agricultural expansion drives deforestation in the Brazilian Amazon, the world's largest tropical forest. A major reform strengthened the enforcement of forest protections, slowing deforestation. But subsequent political realignment weakened these protections, raising questions about the durability of conservation policy.

2.1 Background

Brazil's Constitution and first Forest Code offer protection against deforestation in the form of the "80 percent rule." According to this rule, private landowners in the Legal Amazon must preserve at least 80 percent of their property as forested land. However, weak enforcement limited the effectiveness of these policies. By the early 2000s, deforestation rates had reached alarming levels, prompting a shift in policy (MMA 2013).

In 2004, the Brazilian government introduced the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm), a comprehensive initiative that integrated technology investments, regulatory enforcement, and economic incentives. A key innovation was the satellite-based DETER monitoring system, which enabled real-time detection of forest loss and strengthened enforcement by the Brazilian Environmental Agency (IBAMA) (Assunção et al. 2023). Legal reforms also reinforced conservation efforts. Deforestation on untitled land was reclassified as a felony, empowering authorities to seize assets used in illegal activities, and protected areas were expanded (Soares-Filho et al. 2010, Harding et al. 2021). These policies

under PPCDAm led to a sharp decline in deforestation, which fell by 70 percent between 2006 and 2012.

As priorities shift and economic pressures build, questions remain around long-term sustainability (Crouzeilles et al. 2017, Costa et al. 2025b). In 2012, the New Forest Code weakened regulation by granting amnesty to properties that had been illegally deforested before 2008 (Soares-Filho et al. 2014, Freitas et al. 2018). Enforcement agencies also faced budget cuts. From 2013 to 2016, IBAMA's budget fell sharply, and reductions to field staff limited monitoring capacity (CGU 2016). These shifts, coupled with broader political realignment in favor of agricultural expansion, coincided with a resurgence in deforestation and the reversal of prior conservation gains (Rochedo et al. 2018, Burgess et al. 2025).

2.2 Election and campaign data

We measure political activity in the form of campaign donations. Our primary data source is the universe of formal donations recorded in all legislative, state, and federal elections from 2002 to 2018. We focus on these elections because state and federal governments hold the greatest authority over environmental policy and enforcement. Elections occur every four years. All campaign donations must be registered and publicly disclosed by the Supreme Electoral Court (TSE), ensuring comprehensive coverage of political contributions. We observe donors, candidates, and donation values. The same TSE data record electoral outcomes: total votes and election outcomes for each candidate at the municipality and election-year level.

We leverage firm-level data from the Brazilian tax authority (*Receita Federal*) to identify agricultural donors and candidates, who are most likely to oppose environmental regulations. These data include the full registry of formal firms and their owners. We use this information to construct a panel of all donors who once owned a firm in Brazil, and we link firm owners to their campaign contributions and candidate affiliations. We classify agricultural sector donors as individuals with firms registered in the agricultural sector. We supplement these data by building a panel of candidates who received donations from firm owners, distinguishing between those funded by donors associated with the agricultural sector and those funded by donors from other sectors. Appendix A provides details on the construction of these panel datasets.

Table 1: Individual donors and candidates

	Individuals			Donation to/from	
	Total	Agricultural firm owners	Infractors	Agricultural firm owners	Infractors
	(1)	(2)	(3)	(4)	(5)
Donors	62,987	3,509	3,564	1,488	530
Candidates	5,605	390	274	1,786	1,716

This table describes individual donors and candidates in state and federal elections in the Legal Amazon from 2002 to 2018. Columns 1 to 3 count distinct individuals: all donors or candidates (1), those who own agricultural firms (2), and those with a history of IBAMA environmental infractions (3). Columns 4 and 5 count individuals who gave or received at least one donation to or from agricultural firm owners (4) or infractors (5). The data come from the Supreme Electoral Court (TSE) for donations and from the Brazilian tax authority (*Receita Federal*) for firm ownership.

We restrict our analysis to donations made in elections within the Legal Amazon, ensuring that our estimates reflect political financing patterns relevant to environmental policymaking in the region. Table 1 presents the number of individual donors and candidates in our sample. Column 1 reports the total number of distinct individuals: 62,987 donors and 5,605 candidates. Of these, column 2 shows that 3,509 donors and 390 candidates own agricultural firms. Furthermore, column 4 reports the number of individuals who gave (or received) at least one donation to (or from) agricultural firm owners. We observe that 1,488 donors gave donations to candidates who owned agricultural firms, while 1,786 candidates received donations from donors who owned agricultural firms.

2.3 Enforcement and deforestation data

We also incorporate data on environmental violations from IBAMA, Brazil's environmental enforcement agency, to identify donors and candidates with a history of regulatory infractions. We obtain individual-level infraction and embargo records. Each record identifies the recipient using a partially masked CPF or CNPJ, along with the violation date and municipality. We link these records to donors and candidates to construct infraction histories. Columns 3 and 5 of Table 1 report the number of donors and candidates with at least one infraction. Column 3 shows that 3,564 donors and 274 candidates have a history of IBAMA environmental infractions. Col-

umn 5 shows that 530 donors gave donations to candidates with a history of IBAMA infractions, while 1,716 candidates received donations from donors with a history of IBAMA infractions.

Turning to regulatory outcomes, the Brazilian National Space Research Institute (INPE) produces annual deforestation estimates through the PRODES program. These data draw on Landsat satellite imagery to detect and measure forest loss across the Legal Amazon. We aggregate the PRODES data to the municipality level for 2000 to 2018. We use this annual deforestation data as a descriptive measure of the impacts of regulation over time, including as it evolves from 2004 to 2006 during the implementation of the PPCDAm reform. For each municipality, the reduction in deforestation during this period serves as a proxy for the extent of exposure to the reform.

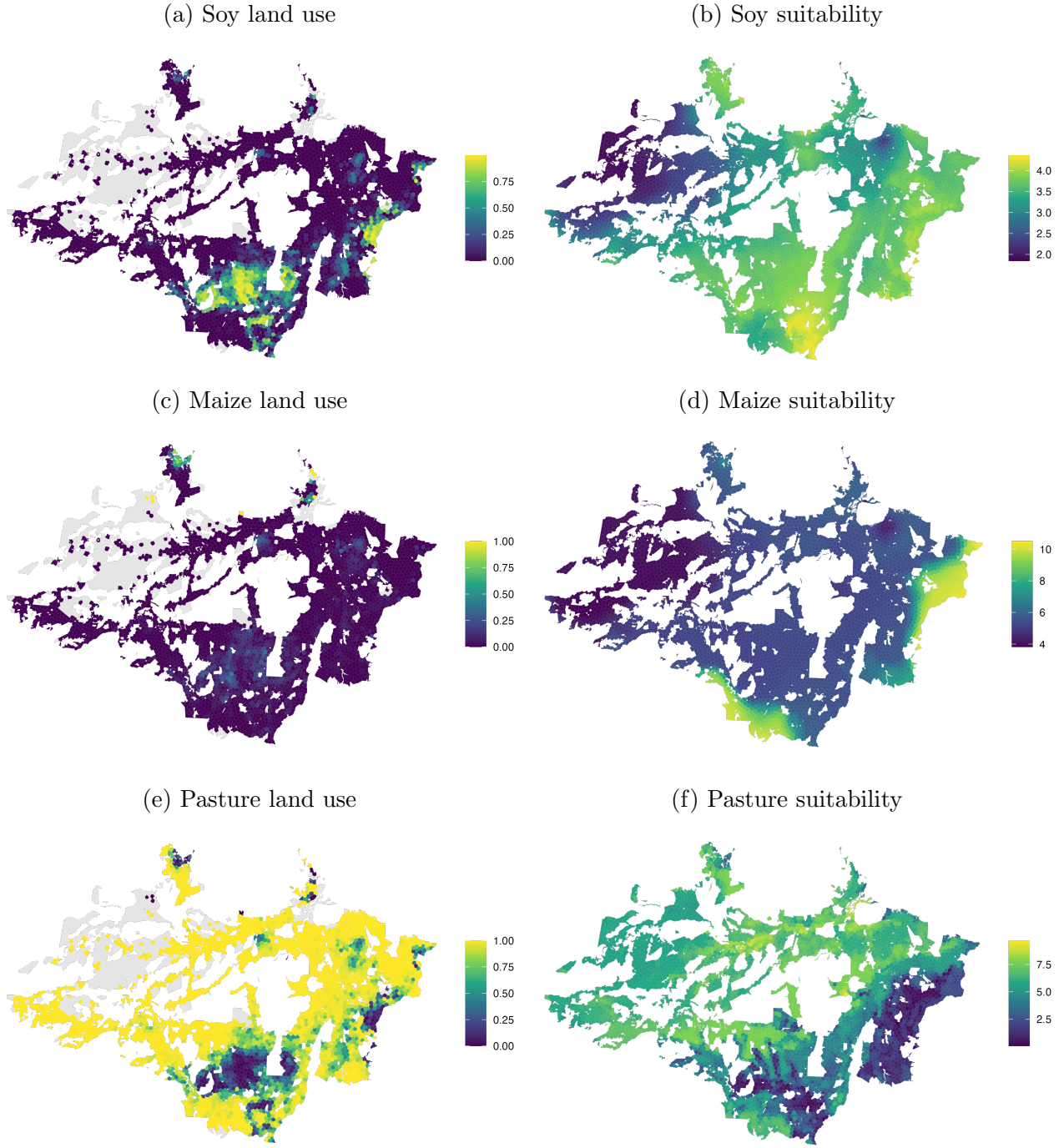
Finally, we use cloud cover data from Assunção et al. (2023) as a measure of exogenous variation in enforcement intensity. The DETER satellite system, which guides IBAMA field operations, cannot detect deforestation beneath persistent cloud cover. We use data on mean municipal cloud cover to capture this plausibly exogenous variation in monitoring capacity across municipalities. Municipalities with high cloud cover are difficult for IBAMA to monitor and thus have lower exposure to the PPCDAm reform. Municipalities with lower cloud cover are easier to monitor and thus more exposed to the reform.

2.4 Land use data

To characterize optimal policy, we compile spatial data on agricultural land use and suitability in the Brazilian Amazon. The data form a spatial panel covering the period from 2000 to 2019. The unit of observation is a 0.25-degree cell, which corresponds to roughly 900 km^2 of land area. The underlying data come from Araujo et al. (2026) and include land allocations to soy, other crops, pasture, and forest from MapBiomas; attainable crop yields for soy and maize in tons per hectare; agro-climatic suitability for pasture; distance to ports; and carbon stocks measured in tons of CO_2 per hectare of land.

Figure 1 presents the data for 2019. In the left column, we map land use shares across agricultural activities. Soy cultivation is concentrated in a few high-yield clusters in the southeast, particularly in Mato Grosso and along the border with the

Figure 1: Agricultural land use and suitability (2019)



Maps of the Brazilian Amazon at the 0.25-degree cell level for 2019. The left column shows the share of agricultural land allocated to each activity. The right column shows attainable crop yields for soy and maize (t/ha), and agro-climatic suitability for pasture. Land use data come from MapBiomas, suitability data come from Araujo et al. (2026).

Cerrado biome. Maize and other crops are concentrated in the same regions but account for a smaller share of agricultural land. Pasture dominates agricultural land use throughout the region, consistent with the extensive cattle ranching that characterizes the Amazon frontier. In the right column, we map crop yields and pasture suitability. Soy and maize yields are highest in the southern and eastern portions of the region, particularly along the "arc of deforestation" where agricultural frontier expansion is concentrated. Pasture suitability is more widespread but still heterogeneous, with the highest values in the south and east.

3 Political Resistance

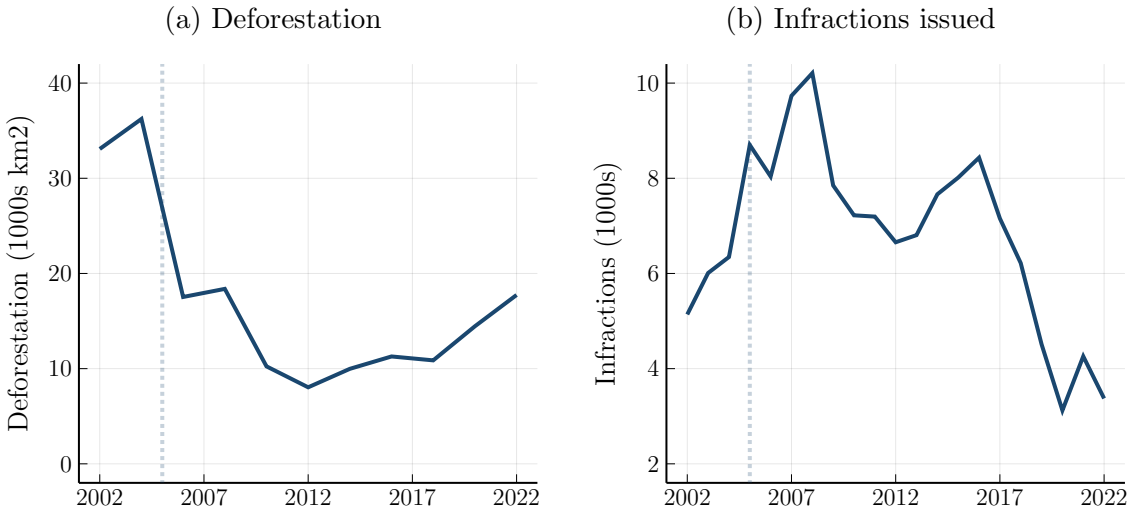
We study political responses to environmental regulation. Stronger enforcement of forest regulations prompted increased campaign donations from agricultural producers. Politicians tied to the agricultural sector raised more funds and won more votes.

3.1 Descriptive evidence

The PPCDAm program strengthened Amazonian protections, increasing enforcement of existing policies and expanding the reach of protected areas. We see the impacts in the data. First, we validate that PPCDAm increased enforcement. Figure 2 shows time series for deforestation and environmental infractions. Deforestation falls after the start of PPCDAm implementation in 2005, while infractions increase. Each pattern is consistent with tighter enforcement of environmental regulation under PPCDAm. Furthermore, we observe an increase in deforestation and a decline in infractions after 2018, which marked a return to lower levels of enforcement.

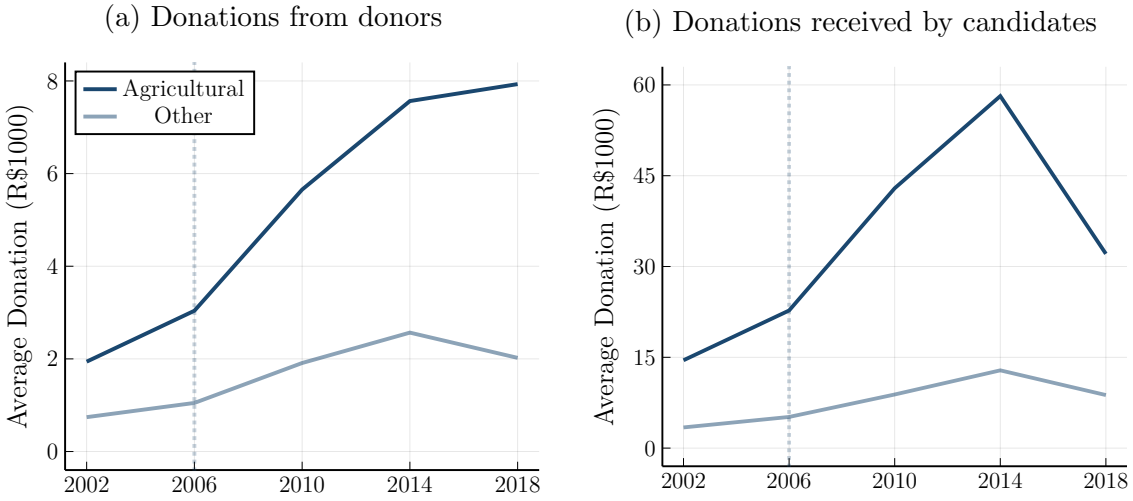
Second, we document political resistance to PPCDAm regulation. Figure 3 plots average campaign donations for donors and candidates from 2002 to 2018 in the Legal Amazon. We include donations to candidates for both state congress (*Deputado Estadual*) and federal congress (*Deputado Federal*). We classify both donors and candidates as agricultural or non-agricultural. To do so, we identified the firms owned by donors and candidates, and we classify these firms according to their CNAE sector codes. We define agricultural firms as those connected to agriculture or food processing, which includes meatpacking, although our results are robust to excluding food processors.

Figure 2: Deforestation and infractions over time



Annual totals for deforestation (left) and environmental infractions (right). The infraction year refers to when authorities formally issued the notice, which may lag the underlying violation. The data come from PRODES for deforestation and from IBAMA for infractions.

Figure 3: Campaign donations over time



Average campaign donations by election year, comparing individuals connected to the agricultural sector (navy) and those connected to other sectors (light blue). Panel (a) shows donations made by donors. Panel (b) shows donations received by candidates. The vertical lines mark 2006, the first election when PPCDAm was fully implemented. The data come from TSE for donations and from *Receita Federal* for sector classification.

Figure 3a contrasts donations from agricultural and non-agricultural donors. We find that contributions from agricultural donors rise sharply after 2006, relative to those from non-agricultural donors. Figure 3b shows the same pattern on the recipient side. Candidates who own agricultural firms receive a large increase in donations after 2006, relative to candidates tied to other sectors. This increase is attenuated in 2018, perhaps reflecting campaign-finance reforms that expanded public funding and reduced reliance on private contributions.[2] Overall, these patterns indicate a disproportionate rise in the average funding of candidates with economic ties to the agricultural sector, which was the most direct target of the PPCDAm reform.

3.2 Identification

We estimate difference-in-differences specifications that formalize the visual comparisons in Figure 3. In particular, we compare agricultural and non-agricultural donations before and after the implementation of PPCDAm. The premise is that agricultural producers are most exposed to the PPCDAm reform, which imposed land use regulation that constrained the scope for agricultural expansion. Non-agricultural producers are less exposed to these constraints on agricultural expansion and thus form a control group. Changes in outcomes for non-agricultural producers therefore capture common exposure to macroeconomic, social, and political fluctuations, but not direct exposure to changes in deforestation policy.

We estimate the effect of increased environmental enforcement on political contributions with the following dynamic specification.

$$y_{imt} = \sum_{\tau} \beta_\tau \, A_i \times \mathbb{1}[t = \tau] + \gamma_{mt} + \varepsilon_{imt}, \tag{1}$$

where y_{imt} is the average donation made by donor i (or received by candidate i) in municipality m and election year t. The indicator $A_i = 1$ if i is an agricultural firm owner, and $\mathbb{1}[t = \tau]$ equals one in election year τ. The coefficients β_τ trace differential outcomes for agricultural versus other sectors by year, relative to 2006. We include municipality-by-election-year fixed effects γ_{mt}. Standard errors are clustered at the individual level (Abadie et al. 2023). We also report average post-2006 effects by

[2] In particular, the creation of the public electoral fund allowed parties and candidates to receive state resources for campaigns. This fund reduced reliance on private contributions, including from the agricultural sector.

pooling across election years.

$$y_{imt} = \beta\, A_i \times \mathbb{1}\big[t > 2006\big] + \alpha_i + \gamma_{mt} + \varepsilon_{imt}, \tag{2}$$

where $\mathbb{1}[t > 2006]$ is an indicator for election years after 2006. We consider alternative fixed-effects sets, including specifications without α_i as a robustness check. Our identifying assumption is that, absent PPCDAm, donations by agricultural owners would have followed parallel trends to those of other owners. We assess this assumption by examining pre-treatment coefficients in the event study.

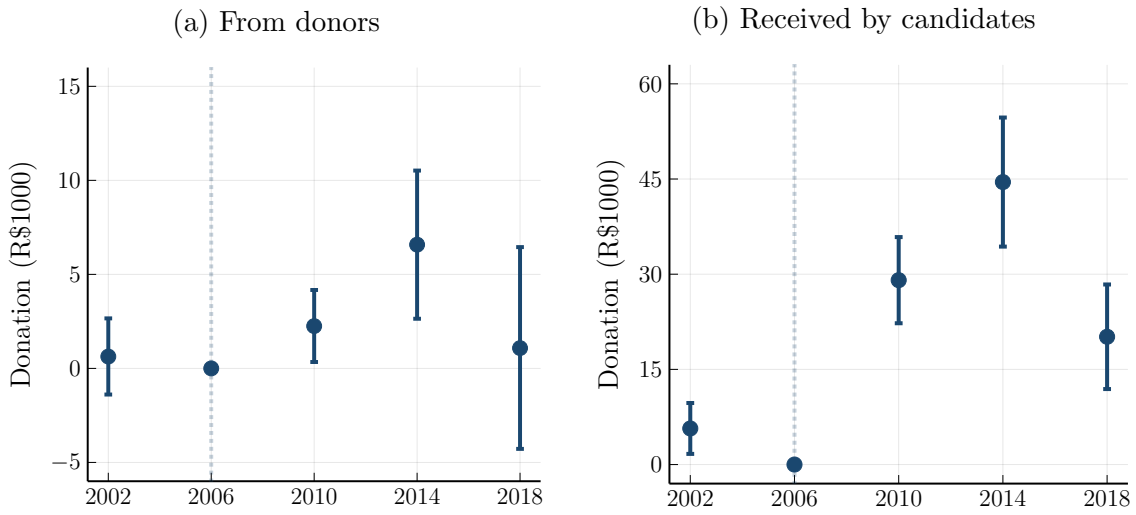
We present alternative specifications that address several potential concerns. First, agricultural firms may differ from other firms in ways correlated with donation behavior. We therefore construct a matched sample, pairing donors (or candidates) by firm characteristics: donations in 2002, firm opening year, participation in the *Simples* tax regime, ownership type (closely held vs. publicly listed), firm size (micro, small, large), and terciles of equity capital. Appendix Table A1 shows that the matched sample indeed improves the balance between agricultural and non-agricultural firms on observed characteristics. We show robustness to matched specifications, which extend equations 1 and 2 to include match-pair fixed effects.

Second, our period of study coincides with a commodity boom and increased competition from China. During this time, agriculture benefited from trade while manufacturing struggled with foreign competition (Costa et al. 2016). We show robustness to triple-difference specifications of the form

$$y_{imt} = \beta_0\, A_i \times \mathbb{1}\big[t > 2006\big] + \beta_1\, A_i \times \mathbb{1}\big[t > 2006\big] \times H_m + \alpha_i + \gamma_{mt} + \varepsilon_{imt}, \tag{3}$$

where H_m captures municipal heterogeneity. We can broaden our scope with a national sample that defines H_m as an indicator for municipalities in the Legal Amazon. The triple-difference specification then compares effects inside vs. outside the Legal Amazon. If rising agricultural incomes nationally were the main driver of our difference-in-difference effects, then our triple-difference estimates would be attenuated because we would observe similar effects outside the Amazon. We can also narrow our scope by defining H_m as cloud cover, which affects enforcement intensity by hindering satellite monitoring (Assunção et al. 2023). The triple-difference specification then compares effects in municipalities with strong vs. weak enforce-

Figure 4: Campaign donations, difference-in-differences



Event-study coefficients β_τ from equation (1) with 2006 as the reference year. Points are estimates, and bars are 95% CIs with individual-level clustering. Panel (a) shows the differential effect on average donations from agricultural firm owners. Panel (b) shows the differential effect on average donations received by candidates who are agricultural firm owners. Estimates based on the matched sample with match-pair and election-year fixed effects. See Appendix Figure A1 for results based on the complete sample.

ment intensity, thereby isolating the impacts of PPCDAm regulation on our political outcomes of interest.

3.3 Donors

Stronger regulation and enforcement brought about by PPCDAm coincide with higher political giving by agricultural donors and higher receipts for agricultural candidates. Figure 4 shows event-study coefficients from equation (1). For donors, Panel (a) shows that agricultural firm owners increased contributions after 2006 relative to owners in other sectors, with a peak in 2014. For candidates, Panel (b) shows that those with agricultural businesses received markedly more funding after 2006.

In Table 2, Panel A reports the difference-in-differences coefficients from equation (2) that capture the average increase in donations by agricultural firm owners after 2006. Columns vary the fixed effects. Column 1 includes municipality and election-year fixed effects. Column 2 replaces these with municipality-by-year fixed effects, flexibly absorbing local-by-year shocks in political contributions. Column 3

Table 2: Donor donations

	(1)	(2)	(3)	(4)
Panel A: Donations to all candidates				
$Ag \times Post$ 2006	5.70***	5.81***	4.02***	3.24***
	(0.91)	(0.97)	(0.95)	(1.19)
Effect as % of mean	237.5	242.1	167.6	135
Observations	36,670	35,195	35,195	7,100
Fixed effects	$m + y$	my	$i + my$	match $+ my$
Panel B: Donations to agricultural candidates				
$Ag \times Post$ 2006	3.05***	3.27***	2.25***	3.36***
	(0.53)	(0.57)	(0.57)	(0.63)
Effect as % of mean	277.5	297.4	204.5	305.7
Observations	36,670	35,195	35,195	7,100
Fixed effects	$m + y$	my	$i + my$	match $+ my$

Difference-in-differences estimates of equation (2). Outcomes are donations made by donors. *Ag* indicates agricultural firm owners; *Post* 2006 equals one for elections after 2006. Outcomes are average donations to all candidates (Panel A) and those to candidates who own agricultural firms (Panel B). Fixed effects vary by column for municipality m, election year y, individual i, municipality-year my, and match pairs. Standard errors are clustered at the individual level. Monetary values are in R$1000.

adds donor fixed effects to the municipality-by-year fixed effects, absorbing donor-specific traits that influence giving. In this specification, the differential increase in donations, β, is identified from within-donor variation over time. Column 4 turns to the matched subsample, including matched-pair and municipality-by-year fixed effects. These estimates are based on within-pair changes in donations over time. Overall, the estimates show that average donations from agricultural donors rose by roughly 130 to 242 percent relative to pre-PPCDAm averages (R$3,240 to R$5,810), all statistically significant at 1 percent.

In Table 2, Panel B shows effects on campaign contributions directed to candidates who are themselves agricultural firm owners. The magnitudes are even larger: agricultural donors increased donations to agricultural candidates by about 204 to 306 percent relative to the pre-PPCDAm baseline (the 2002 election). These results suggest that donors raised contributions in response to stronger environmental enforcement, seeking to expand the sector's political influence in Congress.

Table 3: Donor donations by enforcement intensity

	Heterogeneity (H) by			
	Deforestation reduction		Cloud coverage	
	(1)	(2)	(3)	(4)
$Ag \times Post$ 2006	3.59***	1.37	11.55***	9.77***
	(0.85)	(0.97)	(2.49)	(2.58)
$Ag \times Post$ 2006 $\times H$	1.10***	0.96***	-15.20***	-16.72***
	(0.34)	(0.30)	(4.49)	(4.80)
Observations	34,275	7,400	27,785	5,770
Fixed effects	my	match + my	my	match + my

Triple-difference estimates of equation (3). Outcomes are donations made by donors. H measures local enforcement salience. Columns 1–2: H is the (standardized) municipal drop in deforestation between 2004 and 2006 from PRODES. Columns 3–4: H is (standardized) cloud cover that hampers satellite monitoring from Assunção et al. (2023). Fixed effects vary by column for municipality-year my and match pairs. Standard errors are clustered at the individual level. Monetary values are in R$1000.

To test whether increased donations reflect resistance by the agricultural sector to the government's environmental policy, we examine heterogeneous effects using two measures of enforcement intensity at the municipal level. Table 3 reports estimates from the triple-difference specification in equation (3). In columns 1 and 2, we use the municipal reduction in annual deforestation from 2004 to 2006, measured using PRODES. The triple-interaction coefficients are positive and statistically significant at the 1% level, indicating larger post-2006 increases in donations in municipalities that experienced bigger declines in deforestation over this period (i.e., deforestation in 2004 minus deforestation in 2006).

In Table 3, columns 3 and 4 exploit enforcement variation arising from differential cloud cover. Assunção et al. (2023) shows that cloud cover reduces the DETER system's monitoring capacity, which identifies deforestation hotspots and guides law enforcement. The triple-interaction coefficients are negative and statistically significant at the 1% level, consistent with smaller donation increases in municipalities with higher average cloud cover (lower enforcement) relative to municipalities with less cloud cover (higher enforcement). Taken together, these exercises suggest that agricultural donors increased donations more where PPCDAm was more binding.

Finally, we estimate a triple-difference specification that compares the relative

Table 4: Donor donations in the Legal Amazon

	Sample outside Amazon			
	Rest of Brazil		Neighboring states	
	(1)	(2)	(3)	(4)
$Ag \times Post$ 2006	0.65	0.53	-0.08	-0.33
	(0.55)	(0.75)	(1.43)	(1.16)
$Ag \times Post$ 2006 × Amazon	2.70***	2.31*	3.77**	3.53**
	(0.97)	(1.24)	(1.66)	(1.58)
Observations	367,295	13,045	147,905	9,080
Fixed effects	my	match + my	my	match + my

Triple-difference estimates of equation (3). Outcomes are donations made by donors. *Amazon* indicates municipalities in the Legal Amazon. Coefficients on the triple interaction compare post-2006 changes for agricultural donors inside vs. outside the Amazon. Columns 1–2 include donors from the whole country; columns 3–4 include donors in the Legal Amazon and in neighboring regions – MA, PI, CE, BA, MG, MS, and GO. Fixed effects vary by column for municipality-year my and match pairs. Standard errors are clustered at the individual level. Monetary values are in R$1000.

increase in agricultural-sector campaign donations inside the Legal Amazon with donations from the rest of Brazil. In Table 4, columns 1 and 2 show that while donations from the agricultural sector seem to rise nationally, the point estimates outside the Amazon are small and not statistically different from zero. By contrast, the triple-interaction coefficient is about four times larger and statistically significant, implying that the increase in agricultural donations was concentrated in the Amazon. Columns 3 and 4 show similar results when we restrict the outside-Amazon comparison group to donors in neighboring regions. Altogether, the evidence indicates a post-2006 shift in campaign contributions by agricultural donors concentrated in the Amazon. Appendix Table A2 shows that these results are robust to excluding the Cerrado biome and to restricting the sample to the Cerrado only.

3.4 Candidates

Switching the unit of analysis from donors to candidates, we assess the support received over time by candidates who have ever received donations from the agricultural sector. We estimate a difference-in-differences regression analogous to equation (2), with i indexing candidates. Table 5 reports the estimates. In columns 1 and 2, the

Table 5: Candidate donations

	Total donations		Donations from Ag. firm owners		Donations from Infractors	
	(1)	(2)	(3)	(4)	(5)	(6)
$Ag \times Post$ 2006	32.85***	31.24***	11.08***	11.70***	5.08***	5.75***
	(2.79)	(2.53)	(1.17)	(0.94)	(0.87)	(0.88)
Effect as % of mean	176.6	168	257.8	272.2	141	159.6
Observations	15,660	6,290	15,660	6,290	15,660	6,290
Fixed effects	my	match + my	my	match + my	my	match + my

Difference-in-differences estimates of equation (2). Outcomes are donations received by candidates: total (columns 1–2), from agricultural donors (columns 3–4), and from donors with environmental infractions (columns 5–6). Fixed effects vary by column for municipality-year my and match pairs. Standard errors are clustered at the candidate level. Monetary values are in R$1000.

total campaign funds of candidates who received donations from agricultural donors increased by about 170 percent after 2006, relative to candidates financed by owners in other sectors.

Table 5 also shows changes in the composition of candidate donations, further supporting the growing political influence of the agricultural sector. Not only did the total funds received by agriculture-linked candidates increase, but donor composition shifted as well. Columns 3 and 4 show a roughly 250 percent increase in donations received by these candidates from agricultural donors. Columns 5 and 6 suggest over a 140 percent increase in donations received from donors with a history of environmental infractions. These findings indicate a transformation in both the volume and the sources of political financing.

Another way to measure political support for candidates linked to the agricultural sector is to examine electoral outcomes. Table 6 presents the effects of PPCDAm on these candidates' electoral performance, distinguishing between federal (Panel A) and state (Panel B) congressional elections.[3] In columns 1 and 2, agriculture-linked candidates receive about 70 percent more votes than candidates financed by owners in other sectors. In columns 3 and 4, the probability that agriculture-linked candidates for the federal congress are elected rises by more than 80 percent after 2006. For state

[3] Given differences in vote totals and electoral competition across state and federal races, we analyze them separately.

Table 6: Candidate votes

	Votes		Elected	
	(1)	(2)	(3)	(4)
Panel A: Federal Congress				
$Ag \times Post$ 2006	32.61***	38.75***	0.33***	0.41***
	(4.35)	(13.68)	(0.05)	(0.14)
Effect as % of mean	71.2	84.6	81.6	102.6
Observations	1,378	108	1,378	108
Fixed effects	my	match + my	my	match + my
Panel B: State Congress				
$Ag \times Post$ 2006	10.37***	5.68**	0.30***	-0.02
	(0.92)	(2.74)	(0.03)	(0.14)
Effect as % of mean	87.6	48.0	75.5	-5.5
Observations	4,995	247	4,995	247
Fixed effects	my	match + my	my	match + my

Difference-in-differences estimates of equation (2). Outcomes are thousands of votes received by candidates (columns 1–2) and an indicator for being elected (columns 3–4). Panel A considers candidates running for the Federal Congress (*Deputados Federais*) and Panel B those running for State Congress (*Deputados Estaduais*). Fixed effects vary by column for municipality-year my and match pairs. Standard errors are clustered at the candidate level.

contests, point estimates for election probabilities are similar in sign but become small and statistically indistinguishable from zero in the specification with matched-pair fixed effects. Taken together, these results suggest that PPCDAm increased political support for agriculture-linked candidates: they raised more campaign funds, garnered more votes, and, in federal races, were more likely to be elected.

4 Model

We study a simple model of deforestation under regulation. We introduce political resistance, which can undercut regulation by leading to repeal. We discuss implications for optimal policy.

4.1 Regulation

We study a continuum of heterogeneous forested plots i, which choose to develop agricultural lands $a_i \in \{0, 1\}$. Development generates agricultural value $v_i > 0$, but it requires deforestation that releases carbon emissions $e_i > 0$ at social cost $\kappa > 0$ per ton. The regulator chooses regulation $\tau = \{\tau_i\}_i$ with $\tau_i \geq 0$ to maximize total welfare $W(\tau)$ across plots.

$$W(\tau) = \sum_i w_i(\tau_i)$$

Our analysis will focus on regulation in the form of a tax on agricultural development. For a given plot, welfare $w_i(\tau_i)$ includes agricultural profits, government revenues, and emission damages. Agricultural profits are agricultural values net of taxation.

$$\pi_i(\tau_i) = (v_i - \tau_i)a_i(\tau_i) \tag{4}$$

Profit maximization yields a cutoff rule for development.

$$a_i(\tau_i) = \mathbb{1}[v_i > \tau_i] \tag{5}$$

Taxation raises government revenues $\tau_i a_i(\tau_i)$, and development incurs emission damages $\kappa e_i a_i(\tau_i)$. Combining these expressions, welfare simplifies to agricultural value net of emission damages. The tax is a transfer from producers to the government.

$$\begin{aligned} w_i(\tau_i) &= \pi_i(\tau_i) + \tau_i a_i(\tau_i) - \kappa e_i a_i(\tau_i) \\ &= (v_i - \kappa e_i)a_i(\tau_i) \end{aligned} \tag{6}$$

Result 1. Pigouvian tax $\tau_i^P = \kappa e_i$ maximizes welfare $W(\tau)$ and $w_i(\tau_i)$.

Proof. Plots maximize profits $\pi_i(\tau_i)$. By equations (4) and (6), $\pi_i(\tau_i^P) = w_i(\tau_i^P)$. □

4.2 Resistance

We introduce political resistance, such that taxation is subject to potential repeal. We consider a simple sequential timeline. First, the regulator announces a tax. Second, agricultural producers lobby for repeal in proportion to their tax burden. Heavy taxation has a higher risk of repeal. Third, the tax is repealed with some probability.

Repeal eliminates taxation for all producers, whether they lobby or not. Fourth, production proceeds.

More formally, tax τ is repealed with probability $R(B)$, which is strictly increasing in the total tax burden $B(\tau)$ across plots.

$$R'(B) > 0, \quad B(\tau) = \sum_i b_i(\tau_i) \tag{7}$$

For a given plot, burden $b_i(\tau_i)$ is the agricultural profits lost under taxation.

$$\begin{aligned} b_i(\tau_i) &= \pi_i(0) - \pi_i(\tau_i) \\ &= \min\{v_i, \tau_i\} \end{aligned} \tag{8}$$

The second line follows from equations (4) and (5), by which $\pi_i(0) = v_i$ and $\pi_i(\tau_i) = \max\{v_i - \tau_i, 0\}$. If taxation halts production, then the plot loses agricultural value v_i. If production continues under taxation, then the plot loses tax τ_i.

Result 2. Repeal risk $R(B(\tau))$ is non-decreasing in tax τ_i.

Proof. By equation (8), $b_i(\tau_i)$ is non-decreasing in τ_i. By equations (7), it follows that R and B are non-decreasing in τ_i. □

4.3 Trade-off

The regulator chooses tax τ to maximize feasible welfare $F(\tau)$, which accounts for repeal risk $R(\tau) = R(B(\tau))$.

$$\begin{aligned} F(\tau) &= W(\tau)[1 - R(\tau)] + W(0)R(\tau) \\ &= W(\tau) - [W(\tau) - W(0)]R(\tau) \end{aligned} \tag{9}$$

The first line expresses welfare in expectation of repeal. The second line highlights the tension between maximizing welfare $W(\tau)$ and minimizing repeal risk $R(\tau)$ when $W(\tau) > W(0)$.

Result 3 formalizes the trade-off and shows where it binds. We build intuition by analyzing an emissions tax $\tau_i^E = \delta e_i$, which applies a common tax rate δ on emissions e_i across plots. It will be convenient to define a plot-specific value ratio

$$\eta_i = \frac{v_i}{\kappa e_i}, \tag{10}$$

which measures agricultural value v_i relative to emission damages κe_i. Plots with $\eta_i > 1$ are socially efficient, while those with $\eta_i < 1$ are socially inefficient.

Result 3. (i) For any tax $\tau_i < \tau_i^P$, the regulator trades off improvements in $W(\tau)$ and $R(\tau)$. (ii) For an emissions tax $\tau_i^E = \delta e_i$, the tax rate that maximizes $F(\tau)$ is $\hat{\delta} = \kappa\hat{\eta}$, where value ratio $\hat{\eta}$ satisfies

$$(1-\eta)\,(1-R)\,\bar{e}(\eta)\,h(\eta) = [W(\kappa\eta) - W(0)]\,R'(B)\,\bar{E}(\eta)\,[1-H(\eta)] \tag{11}$$

for $\bar{e}(\eta) = \mathbb{E}[e_i \,|\, \eta_i = \eta]$, density $h(\eta)$, $\bar{E}(\eta) = \mathbb{E}[e_i \,|\, \eta_i > \eta]$, and distribution $H(\eta)$.

Proof. For (i), $\tau_i < \tau_i^P$ faces a trade-off. On one hand, we have $W(\tau) \leq W(\tau^P)$ by result 1. Relative to τ^P, we lose with τ in terms of W. On the other hand, we have $R(\tau) \leq R(\tau^P)$ by result 2. We gain in terms of R.

For (ii), we focus on $\tau_i^E = \delta e_i$. Tax rate δ implies a cutoff η: the marginal plot has (v, e) such that $v = \delta e$, and so $\delta = \kappa\eta$ by definition of η_i. By equation (9), the first-order condition $F'(\delta) = 0$ yields $W'(\delta)(1-R) = [W(\delta) - W(0)]R'(B)B'(\delta)$. By equations (5) and (6), plots with $\eta_i > \eta$ produce and generate welfare $v_i - \kappa e_i = \kappa e_i(\eta_i - 1)$, while those with $\eta_i < \eta$ do not produce and generate zero welfare. That is, $W(\delta) = \kappa\mathbb{E}[e_i(\eta_i - 1)\cdot\mathbb{1}(\eta_i > \eta)]$, which gives $W'(\delta) = \bar{e}(\eta)(1-\eta)h(\eta)$. By equation (8), plots with $\eta_i > \eta$ bear burden δe_i, while those with $\eta_i < \eta$ bear burden $v_i = \kappa e_i \eta_i$. That is, $B(\delta) = \mathbb{E}[e_i \cdot \min(\kappa\eta_i, \delta)]$, which gives $B'(\delta) = \bar{E}(\eta)[1 - H(\eta)]$. Substituting $W'(\delta)$, $B'(\delta)$, and $\delta = \kappa\eta$ into $F'(\delta) = 0$ yields equation (11).[4] □

Corollary 1. The emissions tax rate that maximizes feasible welfare is less than the Pigouvian rate.

Proof. As above, tax rate δ implies a cutoff η with $\delta = \kappa\eta$ for the marginal plot. For $\delta = \kappa$, the cutoff is $\eta = 1$. The left-hand side of equation (11) is zero because $1 - \eta = 0$, while the right-hand side is strictly positive when $W(\kappa\eta) > W(0)$ and $\Pr(\eta_i > 1) > 0$. We thus have $F'(\kappa) < 0$, which implies $\hat{\delta} < \kappa$. □

[4] We define $\tilde{h}$ as the joint density of (η, e), such that $\bar{e}(\eta)h(\eta) = \int_0^\infty e\tilde{h}(\eta, e)de$. Let $\hat{\eta} = \hat{\delta}/\kappa$ denote the cutoff for F-maximizing tax rate $\hat{\delta}$. From $W(\delta) = \kappa\int_{\hat{\eta}}^\infty\int_0^\infty e(\eta-1)\tilde{h}(\eta, e)\,de\,d\eta = \kappa\int_{\hat{\eta}}^\infty(\eta - 1)\bar{e}(\eta)h(\eta)\,d\eta$, the Leibniz integral rule gives $W'(\delta) = (1-\hat{\eta})\bar{e}(\hat{\eta})h(\hat{\eta})$ for $d\hat{\eta}/d\delta = 1/\kappa$. Similarly, $B(\delta) = \kappa\int_0^{\hat{\eta}}\eta\bar{e}(\eta)h(\eta)\,d\eta + \delta\int_{\hat{\eta}}^\infty\bar{e}(\eta)h(\eta)\,d\eta$ gives $B'(\delta) = \int_{\hat{\eta}}^\infty\bar{e}(\eta)h(\eta)\,d\eta = \mathbb{E}[e_i\cdot\mathbb{1}(\eta_i > \hat{\eta})] = \mathbb{E}[e_i \,|\, \eta_i > \hat{\eta}]\cdot\Pr(\eta_i > \hat{\eta}) = \bar{E}(\hat{\eta})[1 - H(\hat{\eta})]$.

For the marginal plot, the emissions tax δe_i is exactly equal to its agricultural value v_i. Denoting this plot's values as (v, e), $v = \delta e$ implies a cutoff $\eta = \frac{\delta}{\kappa}$. Plots with $\eta_i < \eta$ choose not to produce because the tax exceeds their agricultural value.[5] Plots with $\eta_i > \eta$ choose to produce and pay the tax. Equation (11) equates the marginal benefits and costs of raising tax rate δ. The left-hand side is the expected benefit: at the cutoff $\eta_i = \eta$, we deter plots of density $h(\eta)$ with mean emissions $\bar{e}(\eta)$ if the tax survives repeal risk R. The welfare gain is $(1 - \eta)$ per unit of emissions because the value ratio at the cutoff is η. The right-hand side is the expected cost: we raise the burden on inframarginal producers of mass $1 - H(\eta)$ with mean emissions $\bar{E}(\eta)$, raising repeal risk by $R'(B)$. Repeal forfeits welfare gains $W(\kappa\eta) - W(0)$.

Under Pigouvian taxation, the marginal plot yields zero welfare gain but bears a tax burden that raises repeal risk. Thus, shading the emissions tax below the Pigouvian rate improves feasible welfare, trading second-order welfare gains for a first-order reduction in repeal risk. More generally, the trade-off between welfare gains and repeal risk is governed by hazard rate $\frac{h(\eta)}{1-H(\eta)}$, which measures the density of plots at the margin ($\eta_i = \eta$), relative to the mass of inframarginal producers that continue to produce ($\eta_i > \eta$). When the hazard rate is high, the trade-off is weak: an increase in δ induces large exit and large welfare gains, while burdening a limited stock of inframarginal producers. When the hazard rate is low, the trade-off is strong: an increase in δ induces limited exit and limited welfare gains, while burdening a large stock of inframarginal producers.

4.4 Optimal policy

We now characterize the optimal tax τ_i^* that maximizes feasible welfare $F(\tau)$. Result 3 was constrained to a common tax rate δ. Result 4 is constrained only to $\tau_i \geq 0$. The optimal tax combines a Pigouvian tax with a cutoff that targets high-emission, low-value plots.

Result 4. Optimal tax τ_i^* is given by

$$\tau_i^* = \kappa e_i \cdot \mathbb{1}\left[\eta_i < \frac{1}{1+\mu^*}\right], \quad \mu^* = \frac{[W(\tau^*) - W(0)]\, R'(B(\tau^*))}{1 - R(B(\tau^*))} > 0, \tag{12}$$

where μ^* captures the shadow cost of raising the tax.

[5] If $\eta_i < \eta$, then $\frac{v_i}{\kappa e_i} < \frac{\delta}{\kappa}$ by definition of η_i and η. It follows immediately that $v_i < \delta e_i$.

Proof. For $\tau_i \geq 0$, we show that $\tau_i^* \in \{0\} \cup [v_i, \infty)$. For all $\tau_i \in [0, v_i)$, we have $a_i(\tau_i) = 1$, $w_i(\tau_i) = v_i - \kappa e_i$, and $b_i(\tau_i) = \tau_i$ by equations (5), (6), and (8). Development a_i and welfare w_i are invariant, but burden b_i is increasing in τ_i. Thus, all $\tau_i \in (0, v_i)$ are dominated by $\tau_i = 0$. For $\tau_i \geq v_i$, we have $a_i(\tau_i) = w_i(\tau_i) = 0$ and $b_i(\tau_i) = v_i$. Each is invariant in τ_i, and so outcomes are equivalent in this range.

We now establish a cutoff rule for τ_i^*. At the optimum τ^*, we study the choice of τ_i for a given plot i. We can solve the regulator's problem plotwise because each plot is of measure zero. Differentiating equation (9),

$$\left.\frac{\partial F}{\partial \tau_i}\right|_{\tau^*} = w_i'(\tau_i)\,[1 - R(B(\tau^*))] - [W(\tau^*) - W(0)]\,R'(B(\tau^*))\,b_i'(\tau_i).$$

Since w_i and b_i are linear for $\tau_i \in [0, v_i]$, we can rewrite in differences.

$$\Delta F_i = \Delta w_i\,[1 - R(B(\tau^*))] - [W(\tau^*) - W(0)]\,R'(B(\tau^*))\,\Delta b_i,$$

where $\Delta w_i = w_i(v_i) - w_i(0) = \kappa e_i - v_i$ and $\Delta b_i = b_i(v_i) - b_i(0) = v_i$. Because plot i has measure zero, aggregate welfare $W(\tau^*)$ and burden $B(\tau^*)$ remain unaffected and thus are taken as given. On the right-hand side, the first term is the benefit of imposing $\tau_i = v_i$: welfare gains Δw_i net of repeal risk R. The second term is the cost: added burden Δb_i raises repeal risk by R', threatening repeal at cost $W(\tau^*) - W(0)$. The condition $\Delta F_i > 0$ yields a cutoff rule: $\tau_i^* = v_i$ if $\eta_i < \frac{1}{1+\mu^*}$ and $\tau_i^* = 0$ otherwise, where μ^* is as defined in equation (12). For taxed plots, $\eta_i < \frac{1}{1+\mu^*} < 1$ implies $\kappa e_i > v_i$ by definition of η_i. Since $\tau_i \geq v_i$ yields identical outcomes as shown above, we can equivalently write $\tau_i^* = \kappa e_i$ to obtain equation (12). □

Corollary 2. The optimal tax exempts all efficient plots and some inefficient plots.

Proof. Tax exemptions are given by $\tau_i^* = 0$, and efficiency by $\eta_i > 1$. By equation (12), we have $\tau_i^* = 0$ for all efficient plots because $\eta_i > 1$ implies $\eta_i > \frac{1}{1+\mu^*}$ for $\mu^* > 0$. Furthermore, we have $\tau_i^* = 0$ for some inefficient plots because $\eta_i < 1$ and $\eta_i > \frac{1}{1+\mu^*}$ are both satisfied for $\eta_i \in \left(\frac{1}{1+\mu^*}, 1\right)$. This range is non-empty for $\mu^* > 0$. □

Equation (12) specifies a cutoff rule for value ratio η_i. The optimal tax is a targeted policy. First, it exempts efficient plots. These plots are inframarginal: taxation generates no welfare gains and raises the risk of repeal, unambiguously reducing feasible welfare. Second, the optimal tax also exempts some inefficient plots. These

plots are marginal, and the trade-off of result 3 binds: taxation generates welfare gains but also raises the risk of repeal. The cutoff balances this trade-off. For mildly inefficient plots, exemption is optimal because reduced repeal risk outweighs small welfare losses.

The cutoff reflects a common shadow cost μ^* because repeal risk R depends only on aggregate burden B. That is, additional burden raises repeal risk by the same amount, regardless of which plot is burdened. It is this added repeal risk that generates the shadow cost of raising the tax. Equation (12) characterizes μ^* as a fixed point: μ^* depends on τ^* through $W(\tau^*)$, $B(\tau^*)$, $R(B(\tau^*))$, and $R'(B(\tau^*))$, and τ^* depends on μ^* through the cutoff rule. We can solve for this fixed point numerically.

4.5 Targeting

Result 4 characterizes the optimal tax as a policy that targets on value ratio $\eta_i = \frac{v_i}{\kappa e_i}$. Such targeting requires information on both emissions e_i and agricultural values v_i. Land use emissions are observable by satellite, but agricultural values are unobserved, private information. Result 5 decomposes the value of targeting on v_i by comparing the optimal tax to the emissions tax, which targets only on emissions e_i.

Both taxes operate through value ratio η_i. For the emissions tax, the cutoff is $\hat{\eta} = \hat{\delta}/\kappa$ for $\hat{\delta}$ that satisfies equation (11). A plot produces if and only if $\eta_i > \hat{\eta}$. For the optimal tax, the cutoff is $\eta^* = \frac{1}{1+\mu^*}$ as given by equation (12). A plot produces if and only if $\eta_i > \eta^*$. At a common cutoff, the two policies implement the same allocation but differ in the tax burden that they impose. Thus, the difference between these taxes decomposes into differences in burden and cutoff.

Result 5. The feasible welfare gap between the optimal tax and the emissions tax can be decomposed as follows. For $F^*(\eta) = F(\tau^*(\eta))$ and $F^E(\eta) = F(\tau^E(\eta))$,

$$F^*(\eta^*) - F^E(\hat{\eta}) = \underbrace{F^*(\hat{\eta}) - F^E(\hat{\eta})}_{\text{burden}} + \underbrace{F^*(\eta^*) - F^*(\hat{\eta})}_{\text{cutoff}}.$$

The cutoff difference is of second order. The burden difference is

$$F^*(\hat{\eta}) - F^E(\hat{\eta}) \propto \bar{E}(\hat{\eta})\,[1 - H(\hat{\eta})]$$

for inframarginal emissions $\bar{E}(\eta) = \mathbb{E}[e_i \,|\, \eta_i > \eta]$ and distribution $H(\eta)$.

Proof. The decomposition is an identity. For the cutoff difference, $F^*(\eta)$ is maximized at η^*. Since $\frac{dF^*}{d\eta}\big|_{\eta^*} = 0$, the Taylor expansion yields $F^*(\eta^*) - F^*(\hat{\eta}) = O((\hat{\eta} - \eta^*)^2)$. For the burden difference, common cutoff $\hat{\eta}$ implies that both taxes induce the same allocation. Thus, $W(\tau^*(\hat{\eta})) = W(\tau^E(\hat{\eta}))$. By equation (8), plots with $\eta_i < \hat{\eta}$ bear burden v_i under either tax. Plots with $\eta_i > \hat{\eta}$ bear zero under $\tau^*(\hat{\eta})$ and $\hat{\delta} e_i$ under $\tau^E(\hat{\eta})$. The excess burden is

$$B^E(\hat{\eta}) - B^*(\hat{\eta}) = \hat{\delta} \cdot \mathbb{E}[e_i \cdot \mathbb{1}(\eta_i > \hat{\eta})] = \hat{\delta}\, \bar{E}(\hat{\eta})\, [1 - H(\hat{\eta})]$$

for $B^E(\hat{\eta}) = B(\tau^E(\hat{\eta}))$ and $B^*(\hat{\eta}) = B(\tau^*(\hat{\eta}))$. By equation (9), it follows that

$$\begin{aligned} F^*(\hat{\eta}) - F^E(\hat{\eta}) &= [W(\hat{\eta}) - W(0)]\, [R(B^E(\hat{\eta})) - R(B^*(\hat{\eta}))] \\ &= [W(\hat{\eta}) - W(0)]\, R'(B)\, \hat{\delta}\, \bar{E}(\hat{\eta})\, [1 - H(\hat{\eta})]. \end{aligned}$$

The last line holds to first order and is proportional to $\bar{E}(\hat{\eta})\, [1 - H(\hat{\eta})]$. □

The optimal tax conditions on both emissions e_i and agricultural value v_i. The gains from the latter boil down to the magnitude of inframarginal emissions $\bar{E}(\hat{\eta})\, [1 - H(\hat{\eta})]$, which reflects the joint distribution of (e_i, v_i). Intuitively, the optimal tax maximizes feasible welfare by targeting high-emission, low-value plots. If emissions and values are perfectly positively correlated, then high-emission plots are also high-value plots. The emissions tax mistargets by focusing on high-emission plots alone. A uniform tax across plots may even dominate by better avoiding high-value plots. By contrast, if emissions and values are perfectly negatively correlated, then high-emission plots are also low-value plots. The emissions tax approximates the optimal tax by conditioning on emissions alone.

4.6 Transfers

Our framework takes distributional consequences seriously. The standard approach is to maximize aggregate welfare $W(\tau)$ and rely on lump-sum transfers to address distributional concerns. Under this approach, political resistance is irrelevant: losers from regulation are compensated, and the Pigouvian tax τ^P is optimal without modification. We argue that transfers encounter major difficulties in practice.

The first challenge is informational. Compensating producers requires knowing

their private values v_i, which are unobserved. The regulator cannot simultaneously elicit truthful information about costs and implement efficient transfers, as producers have incentives to overstate their losses. Optimal compensation thus faces a fundamental mechanism design problem. Even well-intentioned transfer schemes will systematically over- or under-compensate individual producers.

The second challenge is fiscal. Full compensation requires transfers on the order of total forgone agricultural profits $\sum_i b_i(\tau_i)$. In the Brazilian Amazon, our empirical analysis will suggest losses of hundreds of billions of dollars. Agricultural returns in this region reflect some of the most productive cropland and pasture in the world, and the number of affected producers spans thousands of municipalities. The scope and scale of fiscal transfers would be unprecedented.

The third challenge is institutional. Site-specific transfers would at least partially target smallholders across remote frontier municipalities, relying on administrative capacity that is largely absent at present. Brazil has experimented with payments for ecosystem services through programs such as *Bolsa Verde* and *Bolsa Floresta*, but their coverage and payment levels are small relative to the compensation needed for the political feasibility of full Pigouvian taxation. The limited reach of these programs is itself evidence that transfers are difficult to implement at the relevant scale.

Beyond these implementation challenges, transfers face credibility and coordination problems. Transfer commitments are subject to time inconsistency: future administrations may renege, and indeed Brazilian environmental policy has shifted dramatically across political cycles. Producers who rationally discount such promises will resist regulation despite the promise of future payments. Furthermore, most benefits of forest protection accrue globally through carbon abatement and biodiversity conservation, but there is currently no effective mechanism for large international-to-local transfers. The experience of REDD+ payments illustrates this difficulty, with modest actual payments over the course of decades.

Finally, and most directly, we observe political resistance to environmental regulation in the historical data. The effects we document in Section 3 are direct evidence that compensating transfers are not taking place. Indeed, Brazilian political discourse centers on who bears the costs of regulation, and not on hypothetical compensating transfers. We model and evaluate the direct incidence of regulation on agricultural producers. Regulatory burden drives political resistance, which the regulator must account for. Our quantification takes this constraint at face value.

5 Quantification

We build and estimate a simple empirical model, which we use to quantify the welfare and emissions impacts of politically optimal policy.

5.1 Model

We bring the theoretical model of Section 4 to data by specifying a logit model of agricultural land use. We define our welfare metrics and our policies of interest. Appendix B presents the logit derivations.

Land use

Sites i choose between agricultural development $a_{ij} \in \{0, 1\}$ and forest $f_{ij} \in \{0, 1\}$ for each hectare j, such that $a_{ij} + f_{ij} = 1$. Development implies deforestation. Sites have fixed endowments of n_i hectares of land, and they choose land use to maximize profits subject to idiosyncratic shocks.

$$\max \left\{ v_i - \tau_i + \epsilon^a_{ij},\ \epsilon^f_{ij} \right\}$$

Agricultural use has value v_i, while forests have private value that we normalize to zero. Taxation τ_i discourages agricultural use. Shocks $(\epsilon^a_{ij}, \epsilon^f_{ij})$ are independent and logit-distributed with scale parameter σ, which governs the sensitivity of agricultural development to taxation. When σ is large, the choice is dictated by the shocks and thus unresponsive to taxation.

Logit shocks deliver simple expressions for agricultural development and profits. By the choice probability formula, the logit analogue of development equation (5) is

$$a_i(\tau_i) = n_i \left[\frac{\exp\left(\frac{v_i - \tau_i}{\sigma}\right)}{1 + \exp\left(\frac{v_i - \tau_i}{\sigma}\right)} \right] \tag{13}$$

for agriculture $a_i = \sum_{j \in i} a_{ij}$, forest $f_i = \sum_{j \in i} f_{ij}$, and total $n_i = a_i + f_i$. By the log-sum formula for utility, the logit analogue of profit equation (4) is

$$\pi_i(\tau_i) = \sigma n_i \log \left[1 + \exp\left(\frac{v_i - \tau_i}{\sigma} \right) \right]. \tag{14}$$

The n_i term aggregates across hectares within each site.

Welfare

Our aggregate welfare metrics are as in Section 4. Producer surplus, government revenue, and emissions are

$$\Pi(\tau) = \sum_i \pi_i(\tau_i), \quad G(\tau) = \sum_i \tau_i a_i(\tau_i), \quad E(\tau) = \sum_i e_i a_i(\tau_i).$$

Welfare is the sum of producer surplus and government revenue, net of emission damages at social cost κ per ton of CO_2. We assume perfectly elastic demand for agricultural goods, such that consumer surplus is zero. We evaluate welfare relative to baseline regulation, which we normalize to $\tau_i = 0$, because logit producer surplus is identified only in differences.[6] Denoting these differences as $\Delta X(\tau) = X(\tau) - X(0)$, we have

$$\Delta W(\tau) = \Delta\Pi(\tau) + \Delta G(\tau) - \kappa \Delta E(\tau). \tag{15}$$

The tax burden is given by producer surplus losses.

$$B(\tau) = \Pi(0) - \Pi(\tau) = -\Delta\Pi(\tau)$$

Feasible welfare gains are welfare gains net of repeal risk.

$$\Delta F(\tau) = \Delta W(\tau)[1 - R(B(\tau))] \tag{16}$$

Policy

We compare three tax policies. First, we study the Pigouvian tax.

$$\tau_i^P = \kappa e_i \tag{17}$$

for social cost $\kappa > 0$. By result 1, this policy maximizes welfare gains ΔW. Second, we consider a more general emissions tax.

$$\tau_i^E = \delta e_i \tag{18}$$

[6] This normalization interprets baseline regulation as non-revenue-generating, such that $G(0) = 0$. Indeed, Brazilian policy has centered on monitoring and protection, rather than taxation.

for tax rate $\delta > 0$, which may not be social cost κ. By result 3, this policy involves a trade-off between maximizing welfare gains ΔW and minimizing repeal risk R. Third, we characterize the optimal tax as a fixed point. This logit analogue of result 4 maximizes feasible welfare gains ΔF.

$$\tau_i^* = \max\left\{0,\, \kappa e_i - \frac{\sigma\mu^*}{s_i(\tau_i^*)}\right\}, \quad \mu^* = \frac{\Delta W(\tau^*)\, R'(B(\tau^*))}{1 - R(B(\tau^*))} \tag{19}$$

for forest share $s_i(\tau_i^*) = f_i(\tau_i^*)/n_i$ and shadow cost μ^* of taxation. It taxes sites with high damages κe_i and high forest shares s_i, where s_i is decreasing in agricultural value v_i. That is, the optimal tax combines a Pigouvian tax with shading that targets high-emission, low-value sites.

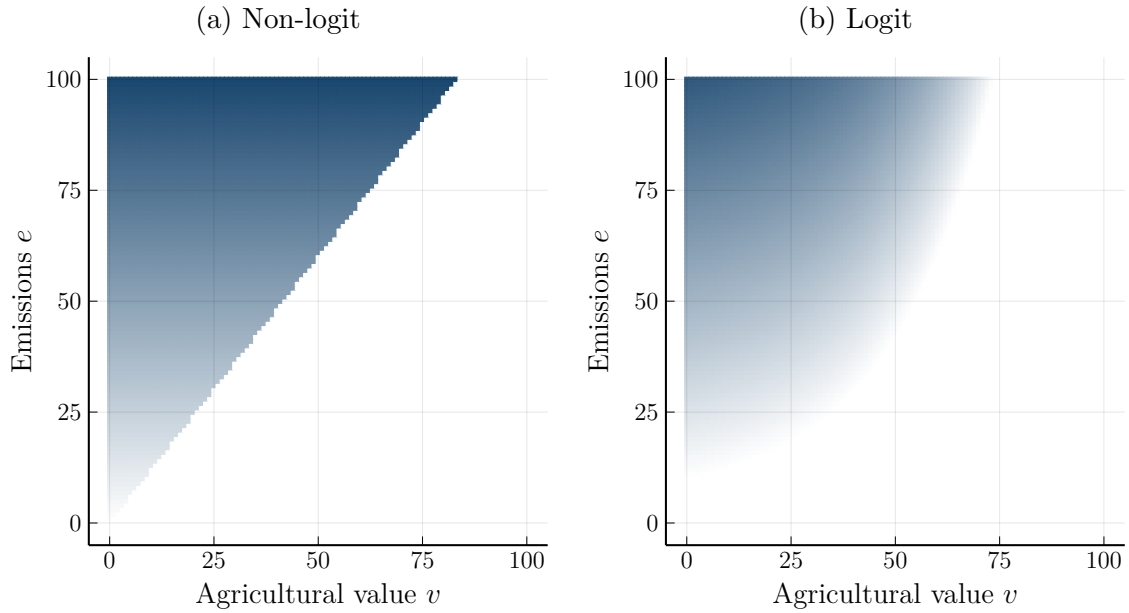
We highlight connections between the optimal tax of equation (19) and its non-logit analogue in result 4, where $\tau_i^* = \kappa e_i \cdot \mathbb{1}[\eta_i < \frac{1}{1+\mu^*}]$. Both expressions start from the Pigouvian rate κe_i and shade down to account for political costs. The non-logit policy does so with a sharp cutoff, exempting all inframarginal and some marginal producers. Inframarginal producers are efficient plots that would bear the tax burden without welfare gains. Exempting them improves feasible welfare with no trade-off. For marginal producers, the political trade-off of result 3 binds and determines where the cutoff is drawn. The logit policy smooths the cutoff, taxing sites at $\kappa e_i - \frac{\sigma\mu^*}{s_i}$. The downward shading increases as forest share s_i falls, and thus as agricultural value v_i rises, until the tax falls to zero. Every producer is marginal to some degree, and so the same trade-off plays out continuously rather than at a sharp cutoff. Figure 5 compares the policies with illustrative values for emissions e and agricultural values v. Both policies impose taxes that increase in e, reflecting that the Pigouvian tax scales with e. For a given e, the non-logit policy applies a sharp cutoff in v, while the logit policy shades continuously in v.

We evaluate each policy by computing counterfactual impacts. For any tax τ_i, we can rewrite equations (13) and (14) as

$$a_i(\tau_i) = n_i \left[\frac{\frac{a_i}{f_i}\exp\left(-\frac{\tau_i}{\sigma}\right)}{1 + \frac{a_i}{f_i}\exp\left(-\frac{\tau_i}{\sigma}\right)}\right], \quad \pi_i(\tau_i) = \sigma n_i \log\left[1 + \frac{a_i}{f_i}\exp\left(-\frac{\tau_i}{\sigma}\right)\right],$$

where we observe agricultural land a_i, forests f_i, and total land n_i as data. Producer surplus, government revenue, and emissions follow from development $a_i(\tau_i)$ and prof-

Figure 5: Pigouvian shading



Illustrative comparison of the optimal tax in the non-logit and logit settings of Sections 4 and 5. Darker blue indicates higher taxation.

its $\pi_i(\tau_i)$. Thus, for counterfactual welfare analysis, we need only to estimate logit parameter σ.

5.2 Estimation

We connect the model to data by constructing agricultural returns and estimating the logit parameter. We do so with agricultural panel data for the Brazilian Amazon from 2001 to 2019.

Agricultural returns

The main empirical challenge is that we do not observe agricultural values v_i, which include both returns r_i and costs c_i.

$$v_i = r_i - c_i \tag{20}$$

Costs are unobserved and encompass a range of economic and institutional barriers, which we cannot directly measure. However, we make progress by constructing a

measure of agricultural return, combining variation across space in agricultural productivity with variation over time in global prices. For $k \in \{\text{soy, maize, pasture}\}$,

$$r_{it} = \sum_k w_i^k y_i^k p_t^k, \tag{21}$$

where crop weights w_i^k aggregate over yields y_i^k and prices p_t^k. Yields are given by attainable crop yields for soy and maize and agro-climatic suitability for pasture.[7] We capitalize annual returns to a present value, assuming a discount rate of 10 percent.

Weights w_i^k determine how local agricultural composition translates global price variation into local profitability shocks. We consider three approaches to weighting. First, we use observed, baseline land-use shares for each site. This measure directly captures local agricultural composition, but it may also partially reflect endogenous land-use choices. In particular, sites' own past land-use decisions may be correlated with unobserved determinants of future agricultural expansion. Second, we use leave-one-out shares based on aggregate shares at the level of Brazilian microregions. We exclude sites' own shares. Third, following Fiszbein (2022), we use fitted shares from a Poisson land allocation model that predicts land-use shares from attainable yields and suitability. These fitted shares reflect expected land allocations given agro-climatic conditions.

Logit parameter

The logit inversion delivers an estimating equation. By equations (13) and (20),

$$\log\left(\frac{a_i}{f_i}\right) = \frac{1}{\sigma}(r_i - c_i)$$

[7] For pasture, we convert agro-climatic suitability into a cardinal measure of productivity. Following Domínguez-Iino (2026), we use the estimated relationship

$$\text{cattle per hectare} = 0.2 + 1.2\,(\text{pasture suitability}).$$

We then convert stocking rates into annual output per hectare, multiplying by 0.5 tons (the average weight of a cattle head) and dividing by 3 (reflecting an average slaughter age of three years). We obtain yields in live cattle weight for slaughter per hectare, which we combine with the price of live cattle to obtain pasture returns. Agricultural prices are from IPEA's farmgate price series for soy, maize, and live cattle (IPEA 2024).

for a given site i, which we define as a 0.25-degree cell. The specification that we take to the data is

$$\log\left(\frac{a_{it}}{f_{it}}\right) = \beta r_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \tag{22}$$

where we add year subscripts t to leverage observed variation over time. We observe agricultural and forested lands (a_{it}, f_{it}), and we have constructed agricultural returns r_{it}. Costs c_{it} are unobserved and given by site fixed effects α_i, year fixed effects γ_t, and the error term ε_{it}. We estimate equation (22) by Poisson pseudo-maximum likelihood (PPML) at the site-year level, weighting sites by land area.[8] The parameter of interest is β, which identifies logit parameter $\sigma = 1/\beta$.

Our main specification is in pooled long differences over a 10-year horizon. For each start year from 2001 to 2009, we compute crop weights based on the start year, we take the long difference to the end year over the 10-year horizon, and we stack these windows into a single dataset. We estimate a common return coefficient across windows, interacting fixed effects with window indicators to allow for window-specific intercepts. Identification thus comes from within-window, cross-site variation in baseline returns. We cluster standard errors at the level of Brazilian microregions to account for autocorrelation in time and space: sites appear in multiple windows, and nearby sites face correlated agricultural conditions.

Table 7 presents our long-difference estimates. Higher profitability leads to expansion of agricultural land at the expense of forest, with consistent estimates across all three approaches to constructing agricultural returns. Column 1 uses observed baseline shares within each site, column 2 uses leave-one-out shares from the microregion, and column 3 uses fitted shares from the Poisson land allocation model. The estimated coefficients lie in a tight range from 0.21 to 0.22 and are statistically significant, with the fitted-share approach yielding the largest point estimate. Appendix Table B1 reports the implied elasticity of agricultural land use to agricultural returns with a mean of 0.67 across sites.

We validate our main results in Appendix B. Figure B1 finds that our long-difference estimates are increasing in the length of the horizon, as is consistent with gradual adjustment of the agricultural frontier. We obtain short-run estimates of less than 0.1 over horizons of less than 5 years, and we obtain longer-run estimates

[8] In exponential form, the specification becomes $a_{it} = f_{it} \cdot \exp(\beta r_{it} + \alpha_i + \gamma_t) \cdot \exp(\varepsilon_{it})$ with f_{it} entering as an offset.

Table 7: Agricultural returns and development

	Crop weights		
	Baseline shares	Leave-one-out shares	Fitted shares
	(1)	(2)	(3)
Agricultural returns	0.21***	0.22***	0.22***
	(0.05)	(0.05)	(0.05)
Observations	37,274	37,274	37,274
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$

PPML estimates of equation (22) in pooled 10-year long differences with data from 2001 to 2019. The dependent variable is agricultural land area with log forest area as an offset. Agricultural returns are in units of $10,000. Each column uses a different approach to constructing the weights in equation (21): observed baseline shares, leave-one-out shares from the microregion, and fitted shares from a Poisson land allocation model. We include site-by-window and year-by-window fixed effects (iw and tw). Standard errors are clustered at the microregion level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

of more than 0.5 over horizons of more than 15 years. Figure B2 shows that our long-difference estimates are largely stable for start years ranging from 2001 to 2009. Our main specification pools across these start years. Table B2 estimates the model at coarser spatial scales: 0.5-degree cells and municipalities, rather than 0.25-degree cells. The estimates range from 0.21 to 0.29 and remain similar across weights. Table B3 shows that our estimates are robust to constructing agricultural returns with three- and five-year moving average prices, rather than with annual prices. Our estimates are consistent across specifications.

5.3 Counterfactuals

We use the estimated model to simulate and evaluate counterfactual policy. We do so for the year 2019, which is the endpoint of our sample. We specify repeal risk R as a probability that increases in aggregate burden $B(\tau) = \sum_i b_i(\tau_i)$ for $b_i(\tau_i) = \pi_i(0) - \pi_i(\tau_i)$. We adopt a simple increasing function.

$$R(B(\tau)) = \min\left\{\frac{B(\tau)}{\lambda}, 1\right\} \tag{23}$$

The probability depends on repeal threshold $\lambda > 0$. Below this level, the risk of repeal increases linearly with burden. Above this level, repeal occurs with certainty. We choose \$500 billion as a conservative baseline value. This figure is roughly one third of average annual GDP in Brazil during the study period. Lower values of λ increase the likelihood of repeal, while higher values decrease the likelihood.

Trade-off

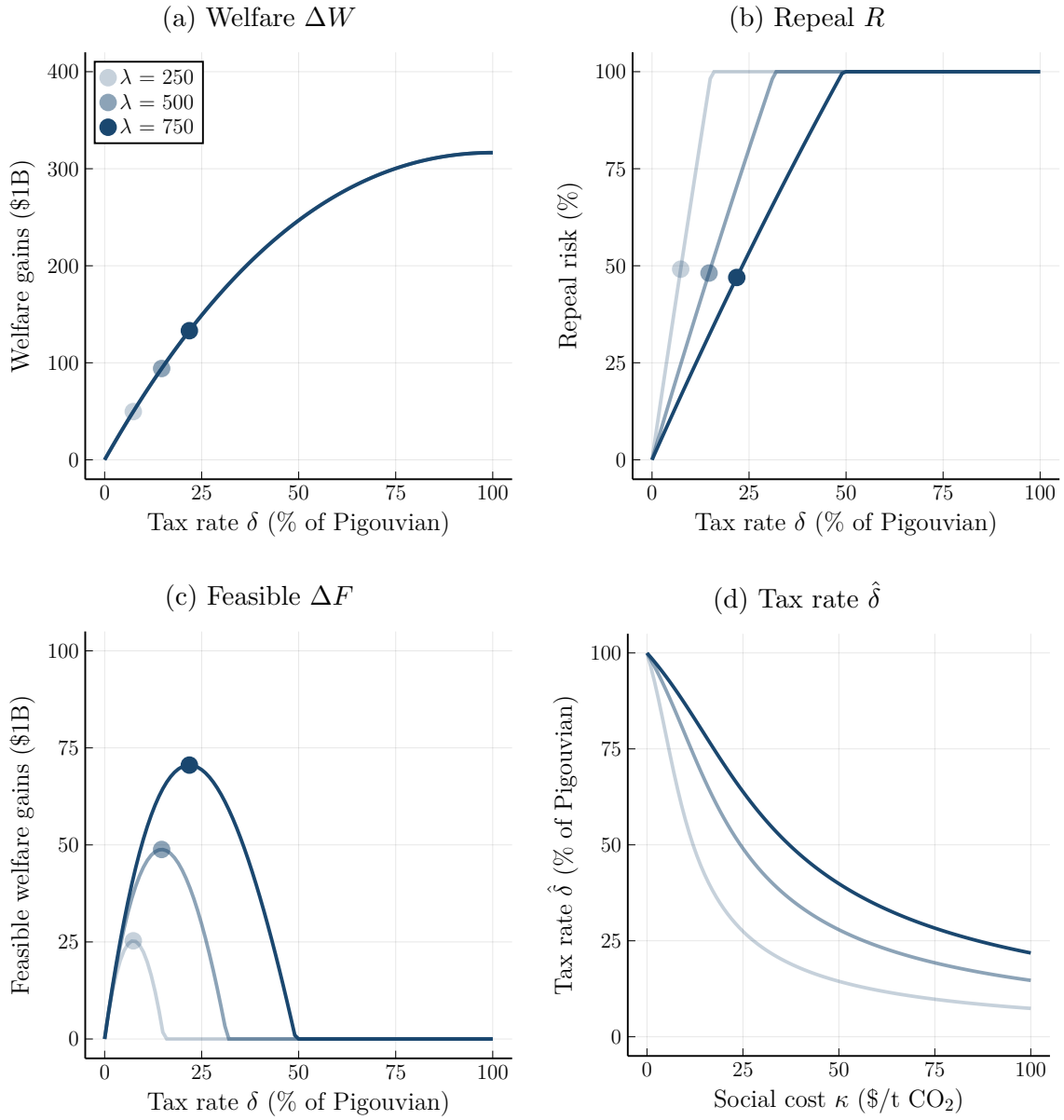
Figure 6 illustrates the trade-off of result 3 for an emissions tax $\tau_i^E = \delta e_i$, as defined by equation (18). We set social cost $\kappa = \$100$ per ton of CO_2, and we vary tax rate δ from \$0 to \$100 per ton of CO_2. We consider repeal threshold values of $\lambda = \$250$, \$500, and \$750 billion.

The trade-off is between welfare gains and repeal risk. On one hand, Figure 6a shows that welfare gains ΔW are increasing in tax rate δ. Higher taxes reduce emissions and raise welfare because social costs exceed the private value of the marginal site for all $\delta \in (0, \kappa)$. Welfare gains are maximized at the Pigouvian rate $\delta = \kappa$, as in result 1.[9] On the other hand, Figure 6b shows that repeal risk R is increasing in δ, as in result 2. Higher taxes impose greater burden on producers and thus raise the probability of repeal. The λ curves shift downward with higher λ values, which imply higher thresholds that reduce repeal risk for any given δ.

Figure 6c shows the implications for feasible welfare gains ΔF. These gains initially rise with the tax rate as welfare gains dominate, but then fall as repeal risk takes over. For each value of λ, a dot marks the tax rate that maximizes feasible welfare gains. This tax rate $\hat{\delta}$ falls short of full Pigouvian taxation, which features maximal welfare gains but also elevated repeal risk. A lower tax rate gives up marginal welfare gains in order to lessen the likelihood of repeal. Figure 6d plots the optimal tax rate across values of (κ, λ). Higher social costs κ make Pigouvian taxation increasingly untenable, as the burden for producers grows with κ. The optimal tax rate thus falls as a share of the Pigouvian rate. By contrast, higher tolerance λ allows for taxation at higher rates.

[9] The λ curves coincide because welfare gains ΔW depend only on tax τ and not on λ.

Figure 6: Trade-off



Counterfactual impacts of emissions tax $\tau_i^E = \delta e_i$ for tax rate δ and repeal parameters $\lambda = \$250$, \$500, and \$750 billion. Panels (a), (b), and (c) set social cost $\kappa = \$100$ per ton of CO_2 and vary tax rate δ. Dots mark the tax rates $\hat{\delta}$ that maximize feasible welfare gains ΔF. Panel (d) varies social cost κ.

Optimal policy

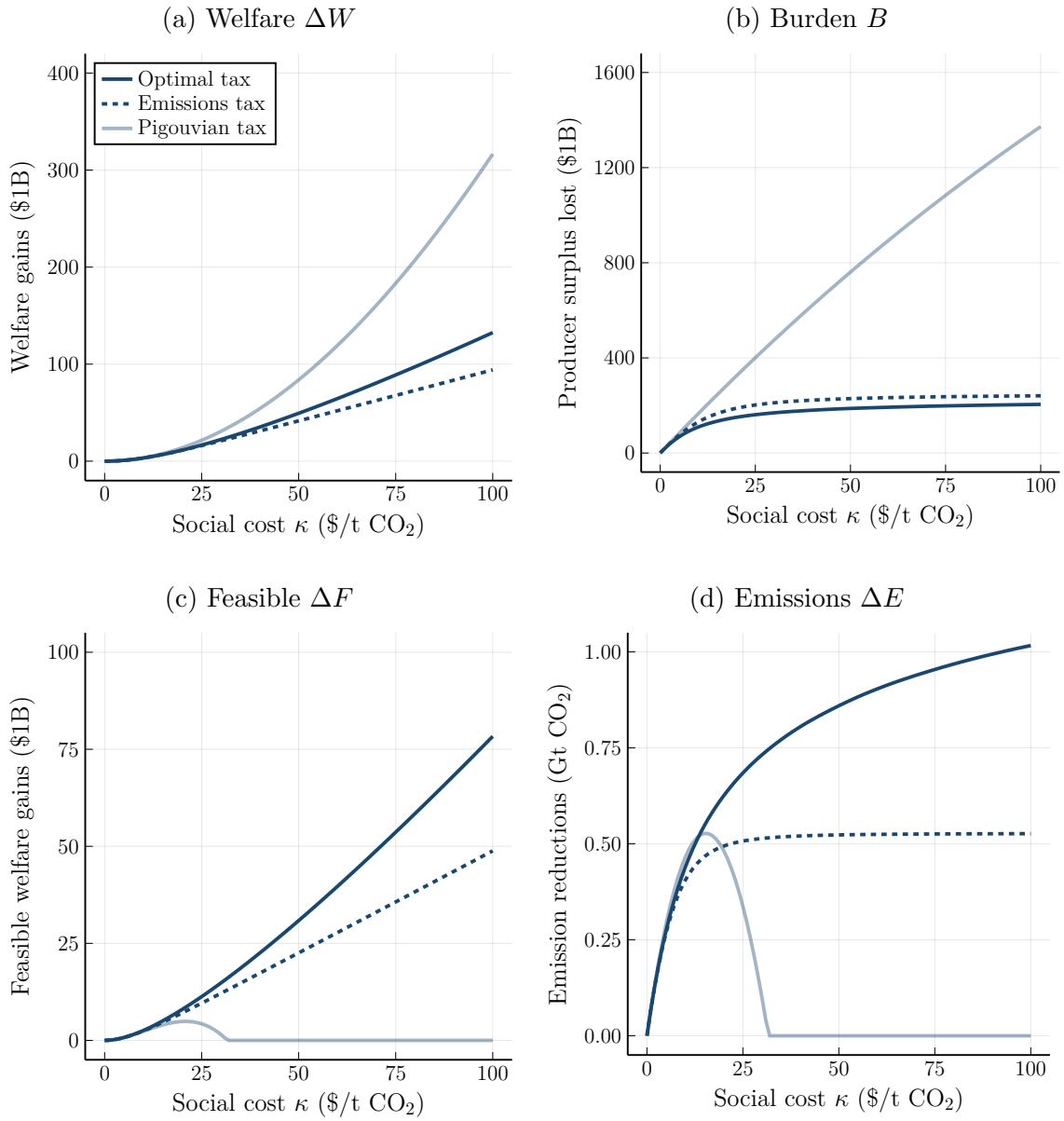
Figure 7 evaluates the optimal tax of result 4. We compare three policies: Pigouvian tax $\tau_i^P = \kappa e_i$ as in equation (17), emissions tax $\tau_i^E = \delta e_i$ as in equation (18), and optimal tax τ_i^* as in equation (19). The Pigouvian tax taxes emissions at social cost κ. The emissions tax targets emissions alone, applying tax rate $\hat{\delta}$ that we choose to maximize feasible welfare gains ΔF. The optimal tax targets both emissions and agricultural values. We set repeal parameter $\lambda = \$500$ billion, and we vary social cost κ from \$0 to \$100 per ton of CO_2.

The Pigouvian tax is worst at navigating the feasible welfare trade-off. Figure 7a shows that welfare gains are increasing in κ, as high social costs raise the welfare gains from taxation. The Pigouvian tax achieves the largest such gains, as in result 1, but only if it survives repeal. The problem is that survival is unlikely. Even at a modest social cost of \$30 per ton of CO_2, Figure 7b shows that Pigouvian taxation imposes producer losses of more than \$500 billion. The resulting political resistance makes repeal inevitable, and we see the net result in Figures 7c and 7d. Pigouvian taxation achieves zero feasible welfare gains, and emission reductions collapse.

The optimal tax is best at navigating the trade-off. Figure 7c shows that the optimal tax achieves the largest feasible welfare gains, as in result 4. The growing gap with other policies highlights the value of targeting. The optimal tax is optimal precisely because it targets on agricultural values, allowing it to extract welfare gains from sites with high emissions but low agricultural values. For these sites, the welfare gains outweigh the tax burden imposed. Targeting results in limited producer losses in Figure 7b, sustaining emission reductions in Figure 7d even when high social costs call for high taxation.

The emissions tax falls between the two. It accounts for the trade-off of result 3 by choosing tax rate $\hat{\delta}$ to maximize feasible welfare gains. But it cannot target on agricultural values, and so it imposes excessive burden on high-value sites. Even so, the emissions tax outperforms Pigouvian taxation because it moderates the tax rate to limit repeal risk. In balancing welfare gains against repeal risk, the emissions tax avoids Pigouvian collapse.

Figure 7: Optimal policy



Counterfactual impacts of three taxes: optimal tax τ^* as in equation (19), emissions tax $\tau^E = \delta e_i$ for tax rate δ, and Pigouvian tax $\tau^P = \kappa e_i$ for social cost κ. We set repeal parameter $\lambda = \$500$ billion and vary social cost κ. The emissions tax applies the tax rate $\hat{\delta}$ that maximizes feasible welfare gains ΔF.

Targeting

Result 5 highlights that the optimal tax targets on agricultural values v_i, which are typically private information and thus unknown to the regulator. We evaluate three additional policies that can be implemented without information on agricultural values. First, we study a revenue tax on observed agricultural returns r_i.

$$\tau_i^R = \delta^R r_i$$

for tax rate $\delta^R > 0$. Second, we consider a flat entry tax on agricultural development.

$$\tau_i^F = \delta^F$$

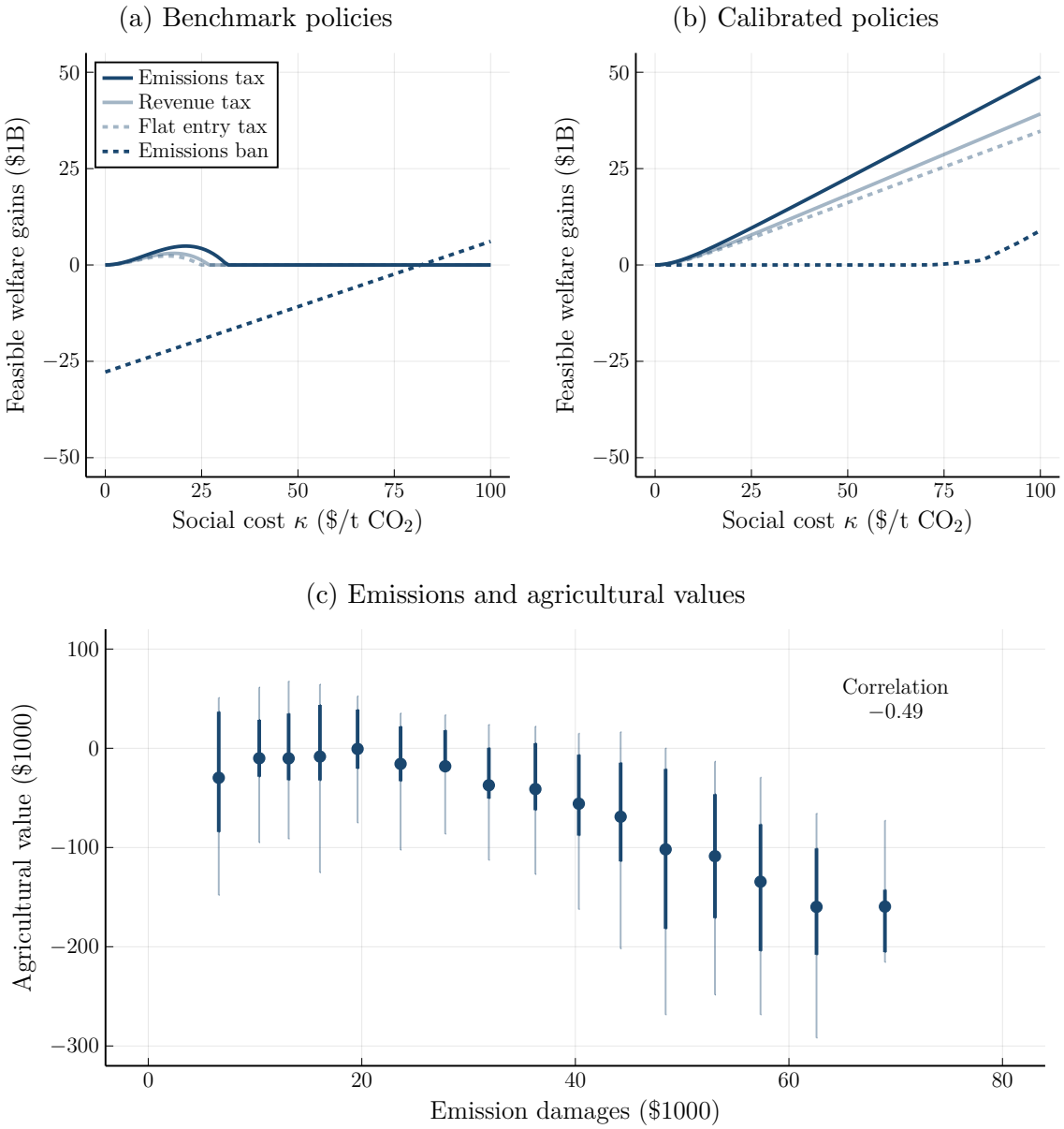
for tax rate $\delta^F > 0$. Third, we simulate an emissions ban that prohibits development on the highest-emitting sites.

$$\tau_i^B = T \cdot \mathbb{1}[e_i > \delta^B]$$

for infinite tax T and emissions cutoff $\delta^B > 0$. We compare these policies to emissions tax $\tau_i^E = \delta e_i$, which also does not condition on agricultural values.

Figure 8a evaluates benchmark policies that do not depend on agricultural values, either for design or for implementation. We set tax rate $\delta = \kappa$ for the emissions tax, $\delta^R = \kappa\bar{e}/\bar{r}$ for the revenue tax, $\delta^F = \kappa\bar{e}$ for the entry tax. We scale by mean emissions and agricultural returns $(\bar{e}, \bar{r})$, such that these taxes coincide for the average site. We set ban cutoff δ^B at the 95th percentile of emissions, banning development on the 5% of sites with the highest emissions. We report feasible welfare impacts. We find that the three taxes collapse at high social cost κ, just as the Pigouvian tax does in Figure 7c. The emissions tax with $\delta = \kappa$ is the same as the Pigouvian tax. It outperforms the other taxes because it differentiates across sites using site-specific emissions. The revenue tax is next because it differentiates using site-specific returns, and the flat entry tax is worst because it is undifferentiated. The problem is that these policies do not account for repeal risk. When social cost is high, they impose large taxes and large producer losses that lead to repeal. The emissions ban is the least effective. It does not generate government revenue, and it still imposes large producer losses that lead to repeal. Thus, the ban achieves positive feasible welfare gains only when social

Figure 8: Targeting



Panels (a) and (b) plot feasible welfare gains ΔF as a function of social cost κ for four policies: emissions tax $\tau_i = \delta e_i$, revenue tax $\tau_i = \delta^R r_i$, flat entry tax $\tau_i = \delta^F$, and an emissions ban that prohibits development on sites with emissions above cutoff δ^B. We set repeal parameter $\lambda = \$500$ billion. Panel (a) sets tax rates at their Pigouvian levels: $\delta = \kappa$, $\delta^R = \kappa\bar{e}/\bar{r}$, and $\delta^F = \kappa\bar{e}$ for mean emissions and revenues $(\bar{e}, \bar{r})$. It sets ban cutoff δ^B at the 95th percentile of emissions. Panel (b) applies the tax rates $(\hat{\delta}, \hat{\delta}^R, \hat{\delta}^F)$ and ban cutoff $\hat{\delta}^B$ that maximize feasible welfare gains ΔF. Panel (c) plots agricultural values v against emission damages κe across bins of sites. We set social cost $\kappa = \$100$ per ton of CO_2. Dots mark bin means, and vertical lines mark interquartile and interdecile ranges. The correlation is between the unbinned values.

costs are high. Otherwise the welfare gains do not outweigh the political costs.

Figure 8b shows that all instruments improve substantially when we calibrate their parameters to maximize feasible welfare gains ΔF. For each policy, we use our empirical model to compute the parameters $(\hat{\delta}, \hat{\delta}^R, \hat{\delta}^F, \hat{\delta}^B)$ that maximize ΔF. Given the calibrated parameters, the regulator can then implement these policies without agricultural values. The emissions tax performs best because it targets site-specific emissions. In accounting for repeal risk, it achieves feasible welfare gains that grow steadily with κ. The revenue tax and flat entry tax capture a substantial share of these gains despite not targeting emissions directly. The emissions ban remains the least effective because it cannot modulate the intensity of regulation. Sites are either banned or not.

Figure 8c illustrates why instruments that target emissions perform relatively well without information on agricultural values. We recover agricultural values by equation (13), which implies $v_i = \sigma \log \frac{a_i}{f_i}$, and we plot them against emission damages for social costs of \$100 per ton of CO_2. We find that sites with high emissions tend to have lower agricultural values, and so taxing emissions achieves welfare gains where producer burden is relatively smaller. This negative correlation explains why the emissions tax compares favorably to the optimal tax in Figure 7c and also to other taxes in Figure 8b. At the same time, the moderate correlation of -0.49 leaves room for improvement by targeting on agricultural values. To target on emissions is to target on agricultural values, but only imperfectly.

6 Conclusion

Environmental regulation can generate political resistance that undermines its own durability. We study this dynamic in the Brazilian Amazon, where conservation policy provoked a political backlash from agricultural producers. Using rich political data, we find that producers responded to environmental reforms by increasing support for pro-agricultural politicians who opposed the reforms. Motivated by this evidence, we develop a model of optimal regulation under the threat of political repeal. We estimate the model with data on land allocation and agricultural returns across the Brazilian Amazon from 2001 to 2019.

Our counterfactual analysis shows that accounting for the risk of repeal fundamentally changes optimal policy. Pigouvian taxation maximizes welfare in isolation

but imposes large burdens on producers, triggering certain repeal and thus achieving little in expectation. Optimal policy softens taxation and further targets sites where emissions reductions are large relative to producer losses. This targeted approach achieves substantial feasible welfare gains where Pigouvian taxation cannot. We emphasize that the design of conservation policy should account for these political constraints.

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APPENDIX

A Political Resistance

A.1 Data construction

We detail the construction of the donor and candidate panel data, which we use to characterize political resistance in Section 3.

Campaign finance and electoral records

We obtain campaign finance records from the Brazilian Supreme Electoral Court (TSE), which publicly discloses all registered donations for every election since 2002. For each donation, the records identify the donor by CPF (individuals) or CNPJ (firms), the candidate by CPF, the donation amount, the office contested, and the state of the election. We use elections from 2002 to 2018, covering five state and federal legislative cycles (2002, 2006, 2010, 2014, and 2018), as well as the four intervening municipal cycles (2004, 2008, 2012, and 2016).

We process the raw files year by year, standardizing variable names and cleaning CPF and CNPJ identifiers. We strip leading zeros, remove null placeholders, and standardize name strings. We retain individual donors (11-digit CPFs) separately from corporate donors (14-digit CNPJs), and we produce a deduplicated list of all distinct individual donors and candidates appearing in any election from 2002 to 2018. Corporate donors are tracked separately and excluded from the main panel.

We also use the TSE records to construct electoral outcomes. For each candidate, we compute total votes received at the municipality level, collapsing across polling zones. We retain the candidate's declared occupation code to flag self-reported agricultural occupations as a supplementary classification.

Firm registry

We link individual donors and candidates to the firm registry of the *Receita Federal do Brasil* (RF), which we extract from the CNPJ database as of 2019. The registry records, for each establishment, the CNPJ identifier, primary CNAE sector code, state and municipality, legal form, firm size (*porte*), registration in the *Simples* regime,

and equity capital. We also use the individual owners' file (*sócios* and *representantes legais*), which identifies each owner by a partially masked CPF and a standardized name.

We classify a firm as agricultural if its CNAE code falls below 1,100,000, which covers primary agriculture, livestock, fishing, and food and beverage processing including meatpacking. For each individual in the owners file, we aggregate characteristics across all firms the individual owns. We assign the modal sector and municipality, and we code a dummy variable for each individual that measures whether they own at least one agricultural firm. Equity capital is categorized into terciles, and firm size is recorded as micro, small, or large.

Name matching

The RF database masks 6 of the 11 CPF digits, preventing direct identifier matching between TSE and RF records. We instead link individuals using a probabilistic name-matching procedure implemented in Python. For each donor and candidate in the TSE, we compare the standardized name against all records in the RF owners file using a fuzzy string algorithm. We use a high similarity score threshold of 0.9 for a match, and we validate candidates using the partial CPF digits where available. For each matched individual, we assign firm characteristics from the RF record: sector classification, municipality, and infraction history. Table A1 reports characteristics of the matched sample.

Environmental infractions

We use publicly available infraction and embargo records from IBAMA. Each record identifies the recipient by name, a partially masked CPF or CNPJ, date, and municipality. We create three merge keys: (1) masked CPF and name, (2) CNPJ (for corporate entities), and (3) name alone. For each individual, we record the year of the first infraction and the year of the first embargo, which represents a more severe enforcement action involving an asset seizure order.

We link infraction records to TSE donors and candidates using all three strategies in sequence, applying each in turn and filling unmatched records with the next strategy. We classify a donor or candidate as an *infractor* if any strategy yields a match, and as an *embargoee* if they appear in the embargo subfile.

Panel construction

We construct separate balanced panels for donors and candidates. For donors, we focus on the five federal and state legislative election years: 2002, 2006, 2010, 2014, and 2018. Starting from the deduplicated individual donor list, we link each donor to the firm registry and infraction records using the name-matching and merging procedures described above. For each donor-year cell, we sum donations made to all candidates, to agricultural candidates, and to candidates with infraction histories. We winsorize donation values at the 99th percentile within year, donor type, and state, where our goal is to limit the influence of extreme values. We then expand to a balanced panel with one observation per donor-election-year pair, filling in zeros for years in which a donor made no recorded donation. Time-invariant characteristics, including sector, municipality, and infraction status, are imputed within each donor identifier to fill gaps created by missing elections.

For candidates, we apply an analogous procedure using all election years from 2002 to 2022 at two-year intervals, totalling 11 periods. For each candidate-year cell, we merge donations received from each donor type (CPF or CNPJ), as well as total votes and election outcomes from the electoral returns. We fill in zeros for non-participation years, and we impute time-invariant characteristics as we do for donors.

We restrict our main analysis to donors and candidates linked to Legal Amazon municipalities. We do so using an official IBGE crosswalk of municipalities to the Legal Amazon perimeter, as defined by Brazilian federal law. We drop observations with missing, zero-padded, or otherwise invalid CPF values. After these restrictions, the final sample contains 62,987 individual donors and 5,605 candidates (Table 1).

A.2 Validation

We include additional validation checks for Section 3.

Matching

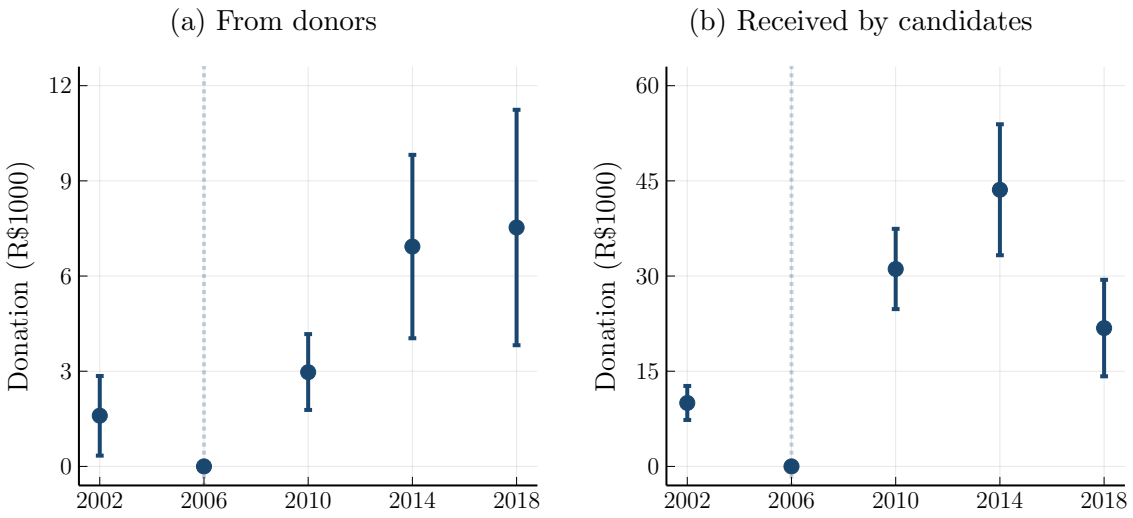
Table A1 assesses covariate balance between agricultural and non-agricultural firms in the full and matched samples. Figure A1 reports difference-in-differences event studies using the complete, unmatched sample, complementing the matched estimates in the main text.

Table A1: Balance of pre-treatment characteristics for agricultural vs. other firms

	Full sample				Matched sample			
	Agri.	Other	Diff.	t-stat	Agri.	Other	Diff.	t-stat
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Donors								
Observations	810	6,524			810	810		
Donation in 2002	2.06	0.52	1.54	2.74	2.06	1.61	0.45	0.68
Firm opening year	2001	2003	-1.75	-4.04	2001	2001	0.18	0.32
Simples tax regime	0.81	0.64	0.17	11.4	0.81	0.80	0.01	0.44
Firm size: micro	0.40	0.54	-0.14	-7.91	0.40	0.40	0	0.10
Firm size: small	0.08	0.09	-0.01	-0.98	0.08	0.10	-0.02	-1.21
LTDA	0.82	0.65	0.17	11.8	0.82	0.83	-0.01	-0.59
Capital: 1st tercile	0.61	0.55	0.07	3.78	0.61	0.58	0.04	1.52
Capital: 3rd tercile	0.34	0.36	-0.02	-1.15	0.34	0.37	-0.03	-1.35
Panel B: Candidates								
Observations	629	2,409			629	629		
Donation in 2002	14.5	3.36	11.2	7.62	14.5	8.82	5.69	3.37
Firm opening year	1999	2002	-2.61	-5.40	1999	1999	0.22	0.34
Simples tax regime	0.84	0.76	0.08	4.85	0.84	0.84	0.01	0.23
Firm size: micro	0.47	0.47	0	0.11	0.47	0.48	-0.01	-0.45
Firm size: small	0.08	0.08	0	0.03	0.08	0.08	-0.01	-0.52
LTDA	0.72	0.60	0.13	6.16	0.72	0.77	-0.05	-1.88
Capital: 1st tercile	0.66	0.62	0.03	1.57	0.66	0.64	0.02	0.71
Capital: 3rd tercile	0.28	0.31	-0.03	-1.60	0.28	0.29	-0.02	-0.62

This table reports balance on pre-treatment firm characteristics for agricultural and other firms. Donations in 2002 are in R$1000. We show the mean difference between agriculture minus other, as well as the t-statistic of the mean equality test. Means for binary variables are proportions. The matched sample pairs agricultural and non-agricultural firms on the characteristics listed. We report two-sided t-tests in columns 4 and 8.

Figure A1: Total donations, difference-in-differences



Event-study coefficients β_τ from (1) with 2006 as the reference year. Points are estimates, and bars are 95% CIs with individual-level clustering. Panel (a) shows the differential effect on average donations from agricultural firm owners. Panel (b) shows the differential effect on average donations received by candidates who are agricultural firm owners. Regressions include municipality-by-election-year.

The Cerrado

Table A2 examines robustness of the national triple-difference to the inclusion and exclusion of the Cerrado biome. A potential concern is that this biome, where soybean production is concentrated, may disproportionately drive the results. To assess this, we report two additional specifications. Columns 1 and 2 exclude municipalities in the Cerrado entirely, comparing agricultural donations inside the Legal Amazon to the rest of the country. The triple-interaction coefficient remains positive and similar in magnitude, indicating that the Amazon-specific increase in agricultural donations does not depend on the Cerrado. Columns 3 and 4 restrict the sample to the Cerrado only, comparing municipalities inside versus outside the Legal Amazon boundary. The point estimates are again consistent with the main findings. Both sets of matched specifications lose statistical significance due to the smaller samples, but the pattern of results is robust across these splits.

Table A2: Donor donations by biome

	No Cerrado		Cerrado only	
	(1)	(2)	(3)	(4)
$Ag \times Post$ 2006	0.34	1.66	1.78***	-0.15
	(0.69)	(1.73)	(0.55)	(3.14)
$Ag \times Post$ 2006 $\times$ Amazon	2.55**	2.10	2.57*	0.72
	(1.23)	(1.49)	(1.42)	(3.05)
Observations	304,175	6,710	63,120	2,760
Fixed effects	my	match + my	my	match + my

Triple-difference estimates of equation (3). Outcomes are donations made by donors. *Amazon* indicates municipalities in the Legal Amazon. Columns 1–2 exclude municipalities in the Cerrado biome; columns 3–4 include only municipalities in the Cerrado. Fixed effects vary by column for municipality-year my and match pairs. Standard errors are clustered at the individual level. Monetary values are in R$1000.

B Politically Optimal Policy

B.1 Model

We present derivations for equations (13), (14), and (19) of the empirical model.

Agricultural development. Each hectare j at site i chooses among two land use options: agriculture with payoff $v_i - \tau_i + \epsilon^a_{ij}$ and forest with payoff ϵ^f_{ij}. Shocks $(\epsilon^a_{ij}, \epsilon^f_{ij})$ are independent and logit-distributed with scale parameter σ. Hectare j chooses agriculture with probability

$$\begin{aligned}
\Pr(a_{ij} = 1) &= \Pr(v_i - \tau_i + \epsilon^a_{ij} > \epsilon^f_{ij}) \\
&= \Pr\left(\frac{\epsilon^f_{ij} - \epsilon^a_{ij}}{\sigma} < \frac{v_i - \tau_i}{\sigma}\right) \\
&= \Lambda\left(\frac{v_i - \tau_i}{\sigma}\right) \\
&= \frac{\exp\left(\frac{v_i - \tau_i}{\sigma}\right)}{1 + \exp\left(\frac{v_i - \tau_i}{\sigma}\right)}.
\end{aligned}$$

The second line collects the shocks and normalizes by scale σ, such that the difference in shocks is of scale one. The third line holds because the difference of two independent logit shocks is logistic with CDF $\Lambda(x) = e^x/(1 + e^x)$. Summing over n_i hectares for each site i, we obtain agricultural development at the site level.

$$a_i(\tau_i) = n_i \left[\frac{\exp\left(\frac{v_i - \tau_i}{\sigma}\right)}{1 + \exp\left(\frac{v_i - \tau_i}{\sigma}\right)}\right]$$

For $a_i(\tau_i) + f_i(\tau_i) = n_i$, it follows that

$$\frac{a_i}{f_i} = \exp\left(\frac{v_i}{\sigma}\right) \tag{24}$$

for $a_i = a_i(0)$ and $f_i = f_i(0)$ as observed in the data. Thus,

$$\exp\left(\frac{v_i - \tau_i}{\sigma}\right) = \frac{a_i}{f_i} \exp\left(-\frac{\tau_i}{\sigma}\right).$$

Substituting, we can rewrite in terms of (a_i, f_i, n_i), policy τ_i, and logit parameter σ.

$$a_i(\tau_i) = n_i \left[\frac{\frac{a_i}{f_i} \exp\left(-\frac{\tau_i}{\sigma}\right)}{1 + \frac{a_i}{f_i} \exp\left(-\frac{\tau_i}{\sigma}\right)} \right]$$

Agricultural profits. For each hectare j, agricultural profits are given by the expected payoff of the optimal land use choice.

$$\begin{aligned}
\mathbb{E}\big[\max\{v_i - \tau_i + \epsilon^a_{ij},\, \epsilon^f_{ij}\}\big] &= \sigma \mathbb{E}\left[\max\left\{\frac{v_i - \tau_i}{\sigma} + \frac{\epsilon^a_{ij}}{\sigma},\, \frac{\epsilon^f_{ij}}{\sigma}\right\}\right] \\
&= \sigma \log\left[\exp\left(\frac{v_i - \tau_i}{\sigma}\right) + \exp(0)\right] \\
&= \sigma \log\left[1 + \exp\left(\frac{v_i - \tau_i}{\sigma}\right)\right]
\end{aligned}$$

The first equality normalizes by scale σ as above. The second line applies the logit log-sum formula. Summing over hectares, we obtain agricultural profits at the site level.

$$\pi_i(\tau_i) = \sigma n_i \log\left[1 + \exp\left(\frac{v_i - \tau_i}{\sigma}\right)\right]$$

Again applying $\exp\left(\frac{v_i - \tau_i}{\sigma}\right) = \frac{a_i}{f_i} \exp\left(-\frac{\tau_i}{\sigma}\right)$, we can rewrite as

$$\pi_i(\tau_i) = \sigma n_i \log\left[1 + \frac{a_i}{f_i} \exp\left(-\frac{\tau_i}{\sigma}\right)\right].$$

Optimal tax. By equation (16), differentiating feasible welfare $\Delta F(\tau)$ with respect to τ_i yields the first-order condition.

$$\frac{\partial \Delta F}{\partial \tau_i} = w_i'(\tau_i)\,[1 - R(B)] - \Delta W(\tau)\, R'(B)\, b_i'(\tau_i) = 0$$

By definition, welfare is $w_i(\tau_i) = \pi_i(\tau_i) + \tau_i a_i(\tau_i) - \kappa e_i a_i(\tau_i)$, and so

$$w_i'(\tau_i) = \pi_i'(\tau_i) + \tau_i a_i'(\tau_i) + a_i(\tau_i) - \kappa e_i a_i'(\tau_i) = (\tau_i - \kappa e_i)\, a_i'(\tau_i).$$

Again by definition, burden is $b_i(\tau_i) = \pi_i(0) - \pi_i(\tau_i)$, and so

$$b_i'(\tau_i) = -\pi_i'(\tau_i) = a_i(\tau_i).$$

In both cases, we obtain the last equalities by applying $\pi_i'(\tau_i) = -a_i(\tau_i)$, which follows from equations (13) and (14). Differentiating equation (13), we have

$$a_i'(\tau_i) = -\frac{1}{\sigma} a_i(\tau_i) s_i(\tau_i)$$

for forest share $s_i(\tau_i) = f_i(\tau_i)/n_i$. The forest share is decreasing in agricultural value v_i by equation (24). Substituting $w_i'(\tau_i)$, $b_i'(\tau_i)$, and $a_i'(\tau_i)$ into the first-order condition,

$$-\frac{1}{\sigma} (\tau_i - \kappa e_i)\, a_i(\tau_i)\, s_i(\tau_i)\, [1 - R(B)] - \Delta W(\tau)\, R'(B)\, a_i(\tau_i) = 0.$$

We solve for τ_i, impose $\tau_i \geq 0$, and define μ to obtain the optimal tax.

$$\tau_i^* = \max\left\{0,\, \kappa e_i - \frac{\sigma \mu^*}{s_i(\tau_i^*)}\right\}, \quad \mu^* = \frac{\Delta W(\tau^*)\, R'(B(\tau^*))}{1 - R(B(\tau^*))}$$

We solve for the fixed point numerically. For a candidate μ, we iterate the left equation to obtain $\tau(\mu)$. We then search over μ to find the μ^* that maximizes $\Delta F(\tau)$.

B.2 Estimation

We describe the construction of agricultural returns, report the supply elasticities implied by our estimates, and validate the main estimates of Table 7.

Agricultural returns

Our third approach to constructing the weights w_i^k in equation (21) uses fitted shares from a land allocation model, following Fiszbein (2022). The idea is to predict each site's agricultural composition from agro-climatic fundamentals, purging the influence of idiosyncratic local factors that may be correlated with unobserved determinants of future development.

Let $k \in \{\text{soy, maize, pasture}\}$ denote agricultural activities. For each activity k, we estimate a Poisson regression of land area on local productivity measures in the baseline year.

$$a_i^k \sim \text{Poisson}(\varphi_i^k), \quad \varphi_i^k = a_i \exp\left(\theta_1^k y_i^{\text{maize}} + \theta_2^k y_i^{\text{soy}} + \theta_3^k y_i^{\text{pasture}}\right),$$

where a_i^k is the land area allocated to activity k in site i, a_i is total agricultural land,

and y_i^k are attainable crop yields for soy and maize and agro-climatic suitability for pasture. The term $\log(a_i)$ enters as an offset, ensuring that predicted allocations scale with total agricultural area. Using the fitted values $\hat{\varphi}_i^k$, we compute the predicted share of activity k in site i as

$$\hat{w}_i^k = \frac{\hat{\varphi}_i^k}{\sum_\ell \hat{\varphi}_i^\ell}.$$

These fitted shares reflect the expected land allocation across activities given local agro-climatic conditions. They are used in place of observed shares when constructing agricultural returns r_{it} in equation (21).

Elasticities

Table B1 reports the distribution of the site-level supply elasticities implied by our long-difference estimates. The elasticity of agricultural land with respect to agricultural returns at site i is $\beta f_i/n_i$, where f_i/n_i is the forest share of total land. Sites with more remaining forest are more responsive to changes in returns, as a given shift in profitability affects a larger share of undeveloped land. The mean elasticity ranges from 0.56 to 0.67 across specifications, and the interquartile ranges span 0.35 to 0.83. These moderate elasticities are consistent with the gradual adjustment of the agricultural frontier over the 10-year horizon of our long-difference estimates.

We report the elasticity of agricultural land area with respect to agricultural returns, which differs from the elasticity of deforestation with respect to agricultural prices. First, deforestation is a flow – the change in the agricultural land stock – which is typically a small fraction of the stock itself. A small proportional change in the stock translates to a large proportional change in the flow, and so a modest stock elasticity can imply a much larger flow elasticity. Second, our elasticities are with respect to composite returns r_{it}, which aggregate across crops. A change in a single crop price translates less than one-for-one into composite returns, and so our elasticities understate the response to individual crop prices. Both forces potentially compress our estimates relative to the deforestation-price elasticities reported in the literature.

Table B1: Elasticities of agricultural land use to agricultural returns

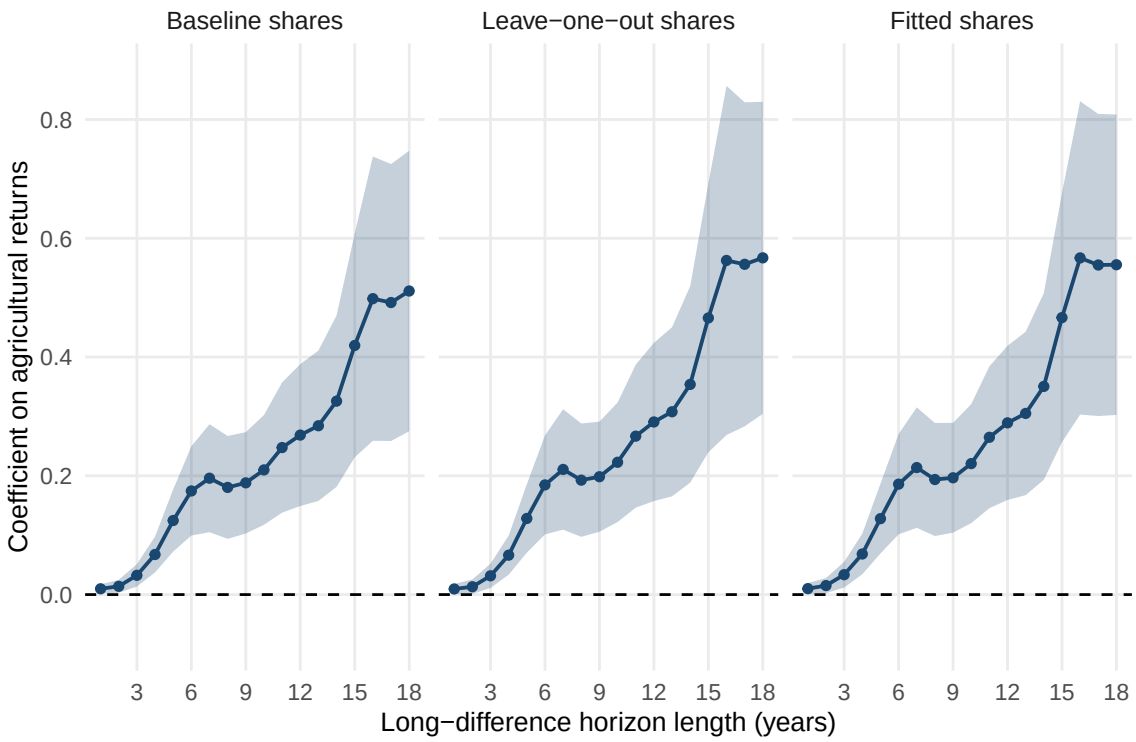
	Crop weights		
	Baseline shares	Leave-one-out shares	Fitted shares
	(1)	(2)	(3)
Mean	0.56	0.67	0.67
	(0.13)	(0.15)	(0.16)
1st percentile	0.15	0.15	0.18
10th percentile	0.27	0.34	0.36
25th percentile	0.35	0.52	0.54
Median	0.54	0.69	0.69
75th percentile	0.75	0.83	0.83
90th percentile	0.89	0.95	0.94
99th percentile	1.08	1.14	1.13

Distribution of site-level elasticities of agricultural land with respect to agricultural returns, as implied by the long-difference estimates of Table 7. Each column uses a different approach to constructing the weights in equation (21): observed baseline shares, leave-one-out shares from the microregion, and fitted shares from a Poisson land allocation model. We compute standard errors of the mean (in parentheses) with the delta method.

Validation

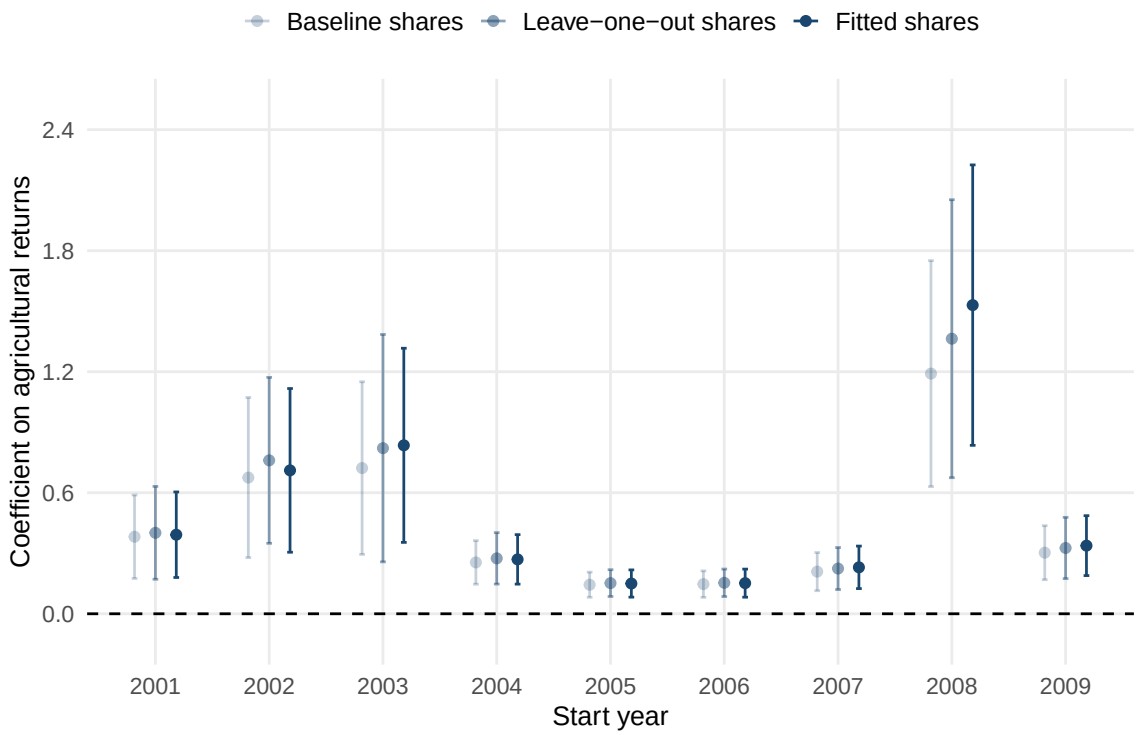
We validate the main estimates of Table 7. Figure B1 shows how the pooled long-difference estimates vary by horizon length, ranging from 1 to 15 years. Figure B2 shows the long-difference estimates for each start year with a 10-year horizon. Table B2 estimates the model at the level of the 0.5-degree cell and the municipality, rather than the 0.25-degree cell. Table B3 estimates the model with moving average prices, rather than contemporaneous yearly prices.

Figure B1: Agricultural returns and development by long-difference horizon



PPML estimates with varying horizon lengths, ranging from 1 to 15 years. For each horizon length, we pool over all available start years. The dependent variable is agricultural land area with log forest area as an offset. Agricultural returns are in units of $10,000. Each panel uses a different approach to constructing the weights in equation (21): observed baseline shares, leave-one-out shares from the microregion, and fitted shares from a Poisson land allocation model. All specifications include site-by-window and year-by-window fixed effects (iw and tw). Standard errors are clustered at the microregion level. The shading marks 95% confidence intervals.

Figure B2: Agricultural returns and development by start year



PPML estimates with varying start years, ranging from 2001 to 2009. Each point corresponds to one 10-year long-difference specification. The dependent variable is agricultural land area with log forest area as an offset. Agricultural returns are in units of \$10,000. Each set of estimates uses a different approach to constructing the weights in equation (21): observed baseline shares, leave-one-out shares from the microregion, and fitted shares from a Poisson land allocation model. All specifications include site and year fixed effects (i and t). Standard errors are clustered at the microregion level. Vertical bars mark 95% confidence intervals.

Table B2: Agricultural returns and development by level of aggregation

	Crop weights		
	Baseline shares	Leave-one-out shares	Fitted shares
	(1)	(2)	(3)
Panel A: 0.25-degree cells			
Agricultural returns	0.21***	0.22***	0.22***
	(0.05)	(0.05)	(0.05)
Observations	37,274	37,274	37,274
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$
Panel B: 0.5-degree cells			
Agricultural returns	0.22***	0.23***	0.23***
	(0.05)	(0.05)	(0.05)
Observations	14,636	14,636	14,636
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$
Panel C: Municipalities			
Agricultural returns	0.28***	0.29***	0.29***
	(0.06)	(0.07)	(0.07)
Observations	7,782	7,782	7,782
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$

PPML estimates at different levels of aggregation. Panel A presents estimates of equation (22) in pooled 10-year long differences with data from 2001 to 2019, aggregated at the level of 0.25-degree cells. It matches our main estimates in Table 7. Panel B aggregates by 0.5-degree cells, and Panel C aggregates by municipality. The dependent variable is agricultural land area with log forest area as an offset. Agricultural returns are in units of $10,000. Each column uses a different approach to constructing the weights in equation (21): observed baseline shares, leave-one-out shares from the microregion, and fitted shares from a Poisson land allocation model. Panel A includes site-by-window and year-by-window fixed effects (iw and tw), while Panel B includes site and year fixed effects (i and t). Standard errors are clustered at the microregion level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B3: Agricultural returns and development with moving average prices

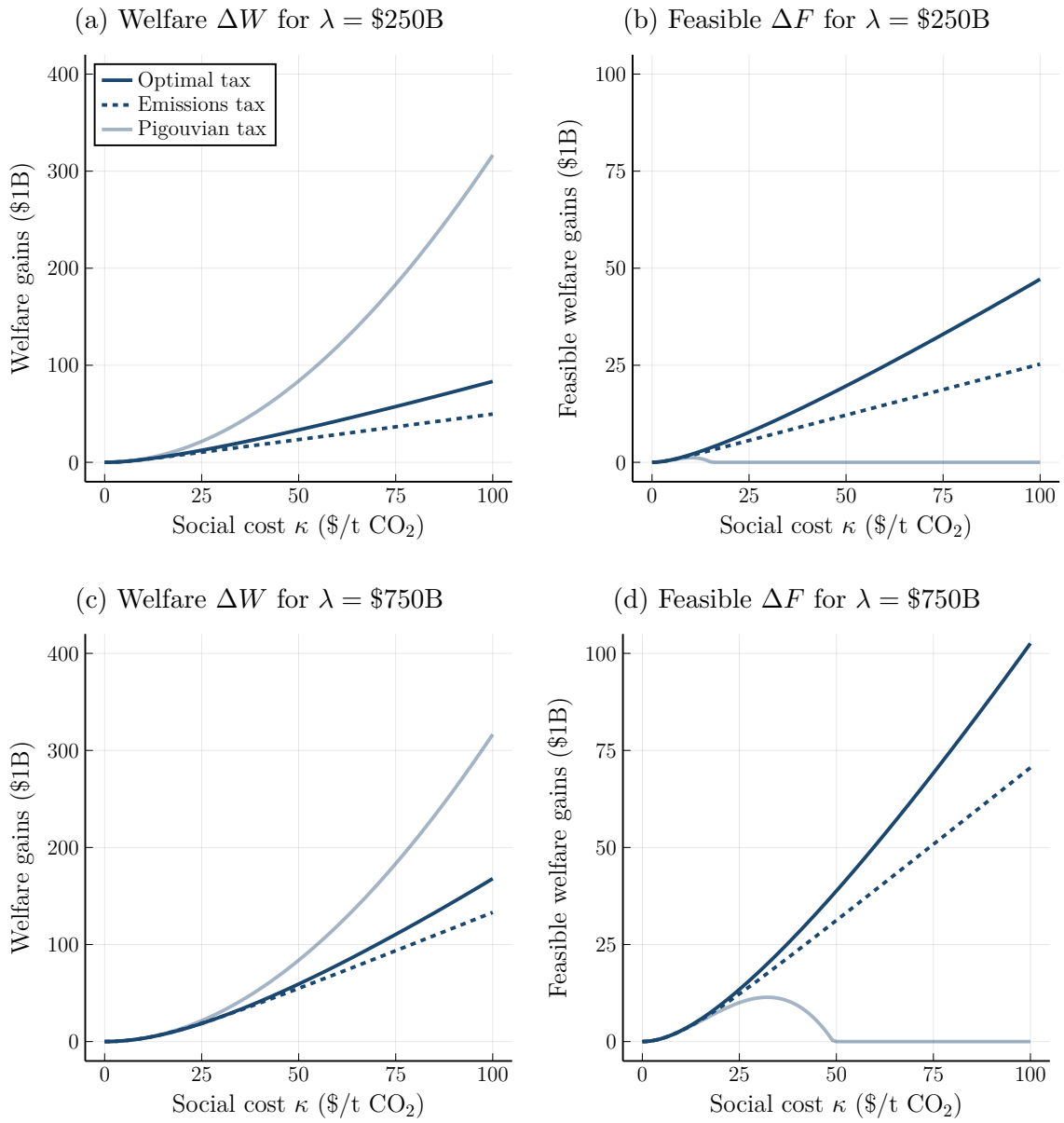
	Crop weights		
	Baseline shares	Leave-one-out shares	Fitted shares
	(1)	(2)	(3)
Panel A: Yearly prices			
Agricultural returns	0.21***	0.22***	0.22***
	(0.05)	(0.05)	(0.05)
Observations	37,274	37,274	37,274
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$
Panel B: 3-year moving-average prices			
Agricultural returns	0.22***	0.24***	0.23***
	(0.05)	(0.06)	(0.05)
Observations	28,966	28,966	28,966
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$
Panel C: 5-year moving-average prices			
Agricultural returns	0.23***	0.25***	0.24***
	(0.05)	(0.06)	(0.06)
Observations	20,690	20,690	20,690
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$

PPML estimates with moving average prices. Panel A presents estimates of equation (22) in pooled 10-year long differences with data from 2001 to 2019, using contemporaneous yearly prices. It matches our main estimates in Table 7. Panel B uses 3-year centered moving average prices, and Panel C uses 5-year centered moving average prices. The dependent variable is agricultural land area with log forest area as an offset. Agricultural returns are in units of $10,000. Each column uses a different approach to constructing the weights in equation (21): observed baseline shares, leave-one-out shares from the microregion, and fitted shares from a Poisson land allocation model. All panels include site-by-window and year-by-window fixed effects (iw and tw). Standard errors are clustered at the microregion level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B.3 Counterfactuals

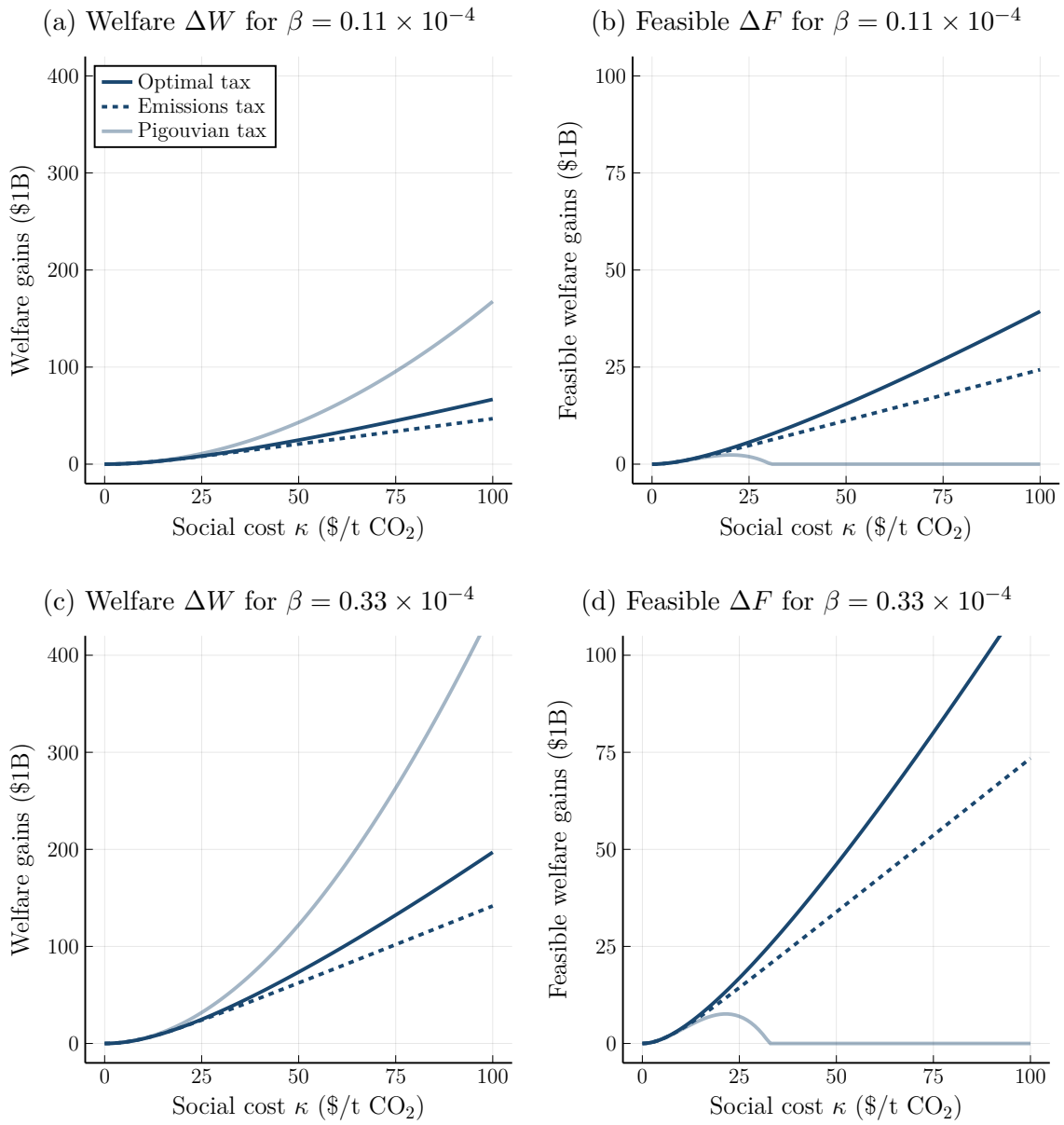
We present additional counterfactual results. Figures B3 and B4 provide sensitivity analysis for the optimal policy of Figure 7.

Figure B3: Optimal policy by λ



Counterfactual impacts of three taxes: optimal tax τ^* as in equation (19), emissions tax $\tau^E = \delta e_i$ for tax rate δ, and Pigouvian tax $\tau^P = \kappa e_i$ for social cost κ. We vary social cost κ. The emissions tax applies the tax rate $\hat{\delta}$ that maximizes feasible welfare gains ΔF. We set repeal parameter $\lambda = \$250$ and \$750 billion. Figure 7 uses our baseline value of $\lambda = \$500$ billion.

Figure B4: Optimal policy by β



Counterfactual impacts of three taxes: optimal tax τ^* as in equation (19), emissions tax $\tau^E = \delta e_i$ for tax rate δ, and Pigouvian tax $\tau^P = \kappa e_i$ for social cost κ. We set repeal parameter $\lambda = \$500$ billion and vary social cost κ. The emissions tax applies the tax rate $\hat{\delta}$ that maximizes feasible welfare gains ΔF. We set the estimated coefficient of equation (22) to $\beta = 0.11 \times 10^{-4}$ and 0.33×10^{-4}. Figure 7 uses our baseline value of $\beta = 0.22 \times 10^{-4}$.